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Link to Module Completion

https://academy.hackthebox.com/achievement/2384006/18

Introduction

I commenced the Linux Fundamentals course and thought about how important it is to work with this OS in general, especially in the field of cybersecurity. The lesson started with the definition of Linux, where it came from, and how widely it was used on different kinds of devices—from single-person workstations and enterprise servers to embedded electronics and smartphones.

Another thing that the course addressed was the principal structure of Linux, which included a timeline of its development, its core architecture, the organization of its directories, and its overall philosophies. Understanding these basics is a prerequisite for all security professionals because they are the very skills that will allow them to navigate and control Linux environments fluently. The first part of the course, thus, set up the following lessons by already stating the most important theories and components that determine the workings of Linux.

Linux Distributions

Linux distributions, or distros as they are commonly known, are the operating systems which consist of the Linux kernel. They have a very wide range of applications, which include running hardware of all kinds, from enterprise servers and specialized devices to personal laptops and mobile phones. It is possible to see the distributions as separate divisions or local branches of one big company, each of them acting according to the needs of particular users or the requirements of specific markets. They all have the same core staff (system components), the same management layout (architecture), and the same guiding principles (philosophy), but every distro still presents its own specialized inventory and amenities (software bundles and settings), which make the user experience for different purposes—all despite being under the collective identity and standards of Linux. Each and every distribution has its uniqueness marked by its unique characteristics, utilities, and management tools.

Introduction to Shell

I got to know that there’s no other way around than to master the Linux shell because most of the servers in the world run the Linux OS. It was a surprise to me that among the reasons why Linux systems are so widely used, one is that Linux is a more stable OS than Windows, particularly in the case of web servers. I came to the conclusion that to manage this OS properly I need to know and be skilled in its most important part: the Shell.

Prompt Description

I gained the understanding of the context that the bash prompt is pretty easy to interpret since generally it shows one’s username, the computer's hostname, and one’s working directory. I can say this line of text is a sign that the system is ready to take my input. I also got to know that the prompt is on the new line with the cursor right after it showing that the system is expecting me to type a command.

Getting Help

I learned that I will run into tools with optional parameters that are still not committed to memory or software that I am completely unfamiliar with, thus it is important to be able to learn about them on one's own. I found that the main ways of doing this are by referring to man pages and using help functions. Additionally, I had the understanding that it is always a good thing to get acquainted with a particular tool/resource before actually using it.

System Information

I discovered that in case of my interaction with several Linux systems, I should have deep knowledge of their internal structure which includes system specs, processes running, network

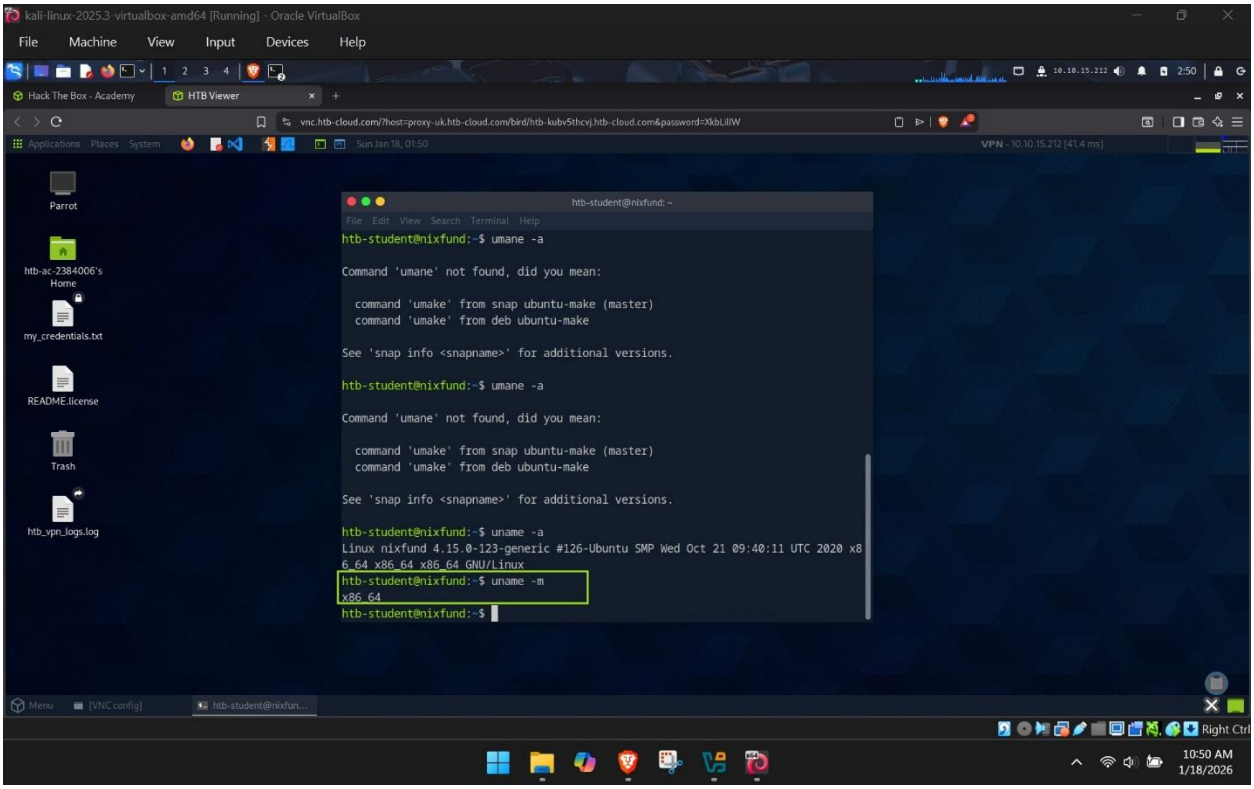
settings, user settings, and directory structures. I could identify a specific collection of core tools, the majority of which are installed already, that would efficiently help me gather this information. Additionally, I recognized that this know-how is not only a requirement for routine admin activities but rather it is the basic requirement for assessing security configurations, detecting weaknesses, and risk reduction strategies within the Linux world.

Questions

1. Find out the machine hardware name and submit it as the answer.

   Answer: x86_64

   This is shown in the image below:

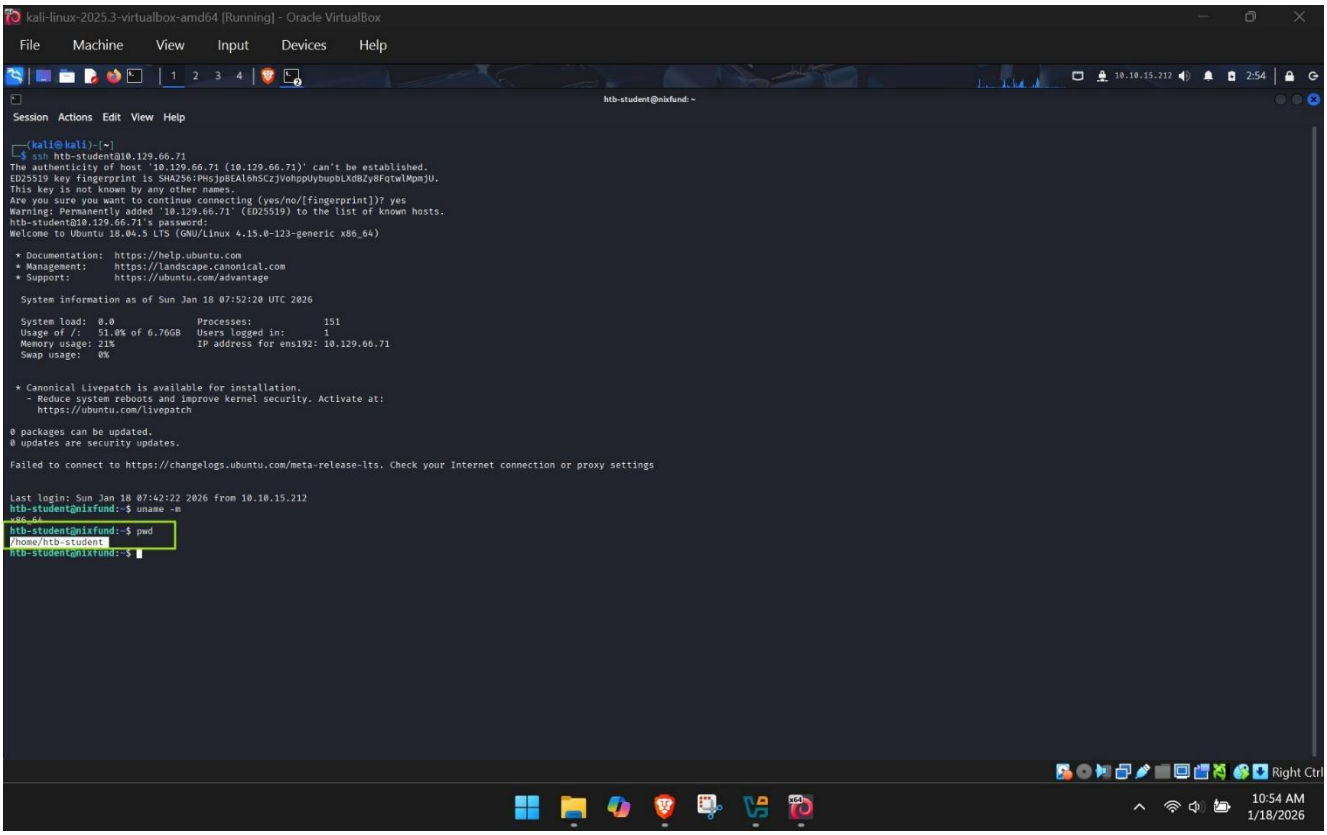


I ran the command uname -m to get the answer.

2. What is the path to htb-student's home directory?

   Answer: /home/htb-student

   This is shown in the image below:

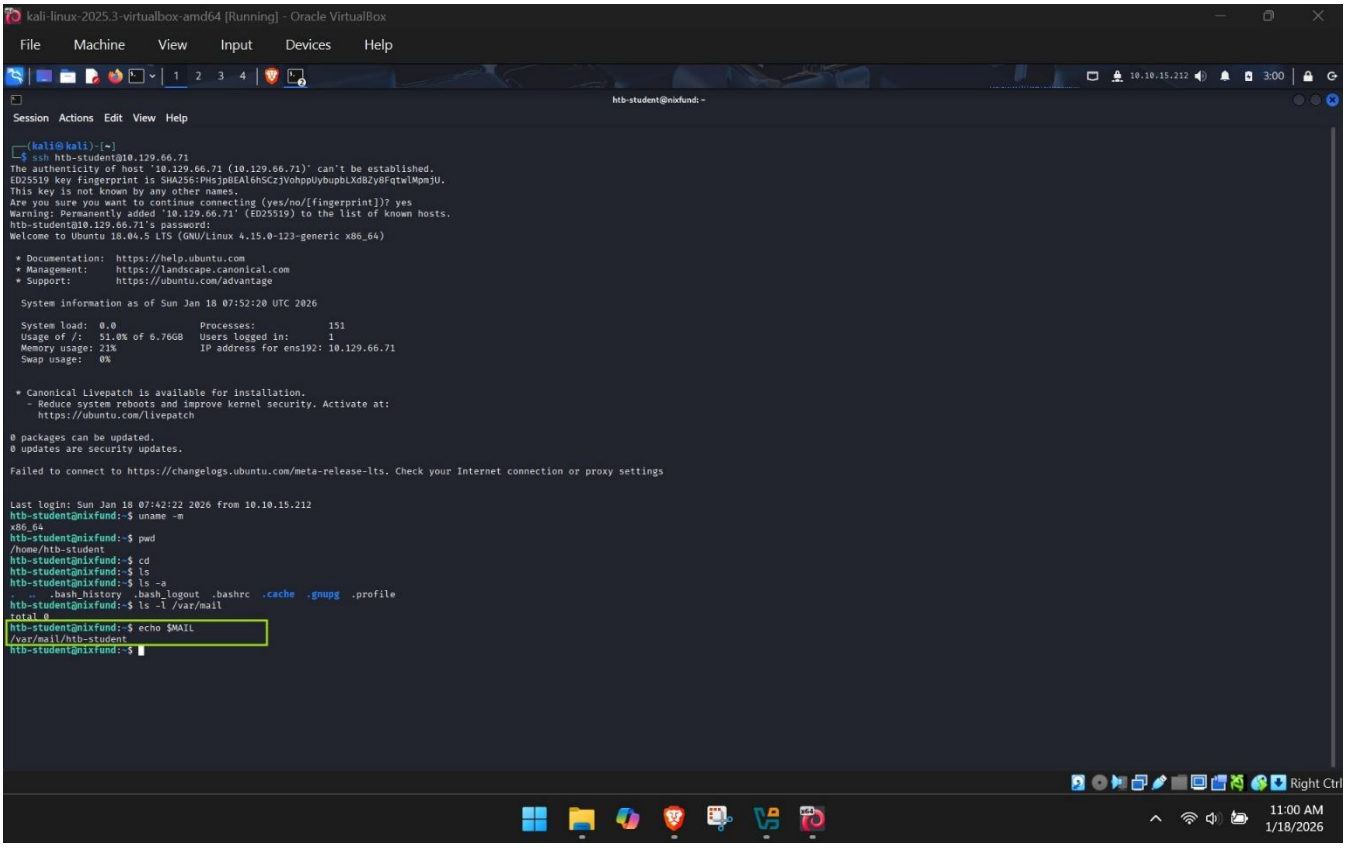


The command I ran was pwd which shows the current directory one is in.

3. What is the path to the htb-student's mail?

   Answer: /var/mail/htb-student

   This is shown in the image below:

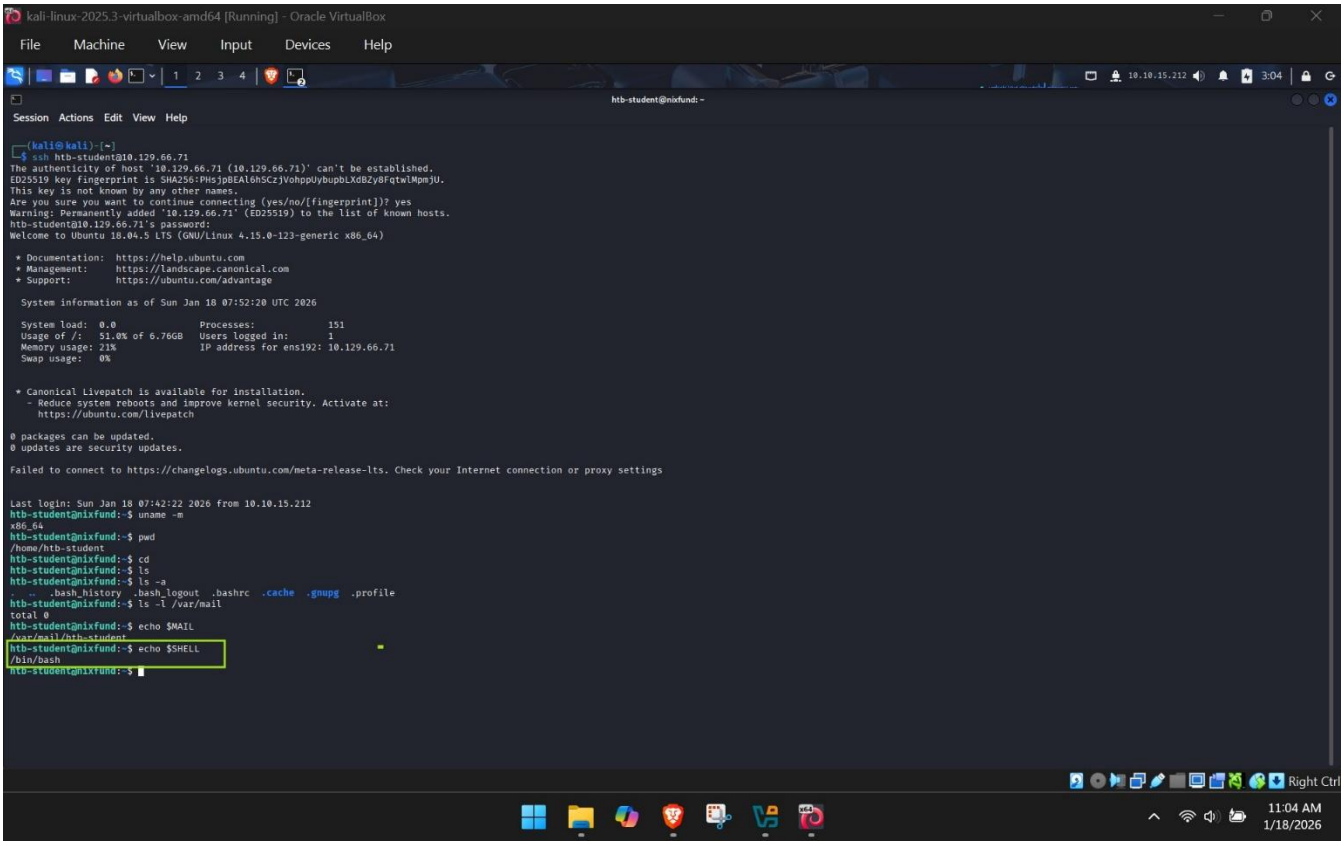


Echo $MAIL is the command I ran to get he actual path to the student's mail.

4. Which shell is specified for the htb-student user?

   Answer: /bin/bash

   This is shown in the image below:

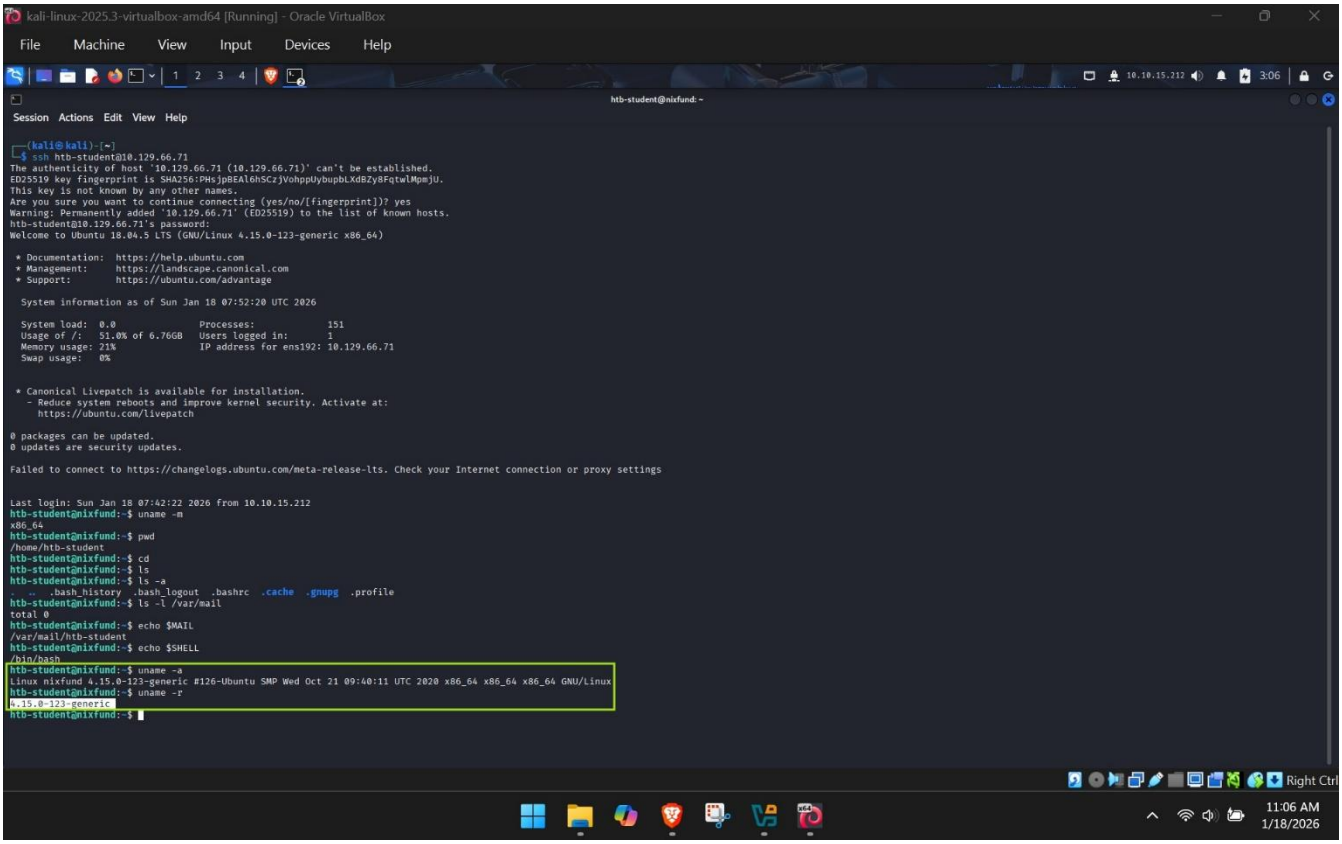


I ran the exact command as the previous question, only that I changed the MAIL to SHELL.

5. Which kernel release is installed on the system? (Format: 1.22.3)

   Answer: 4.15.0

   This is shown in the image below:

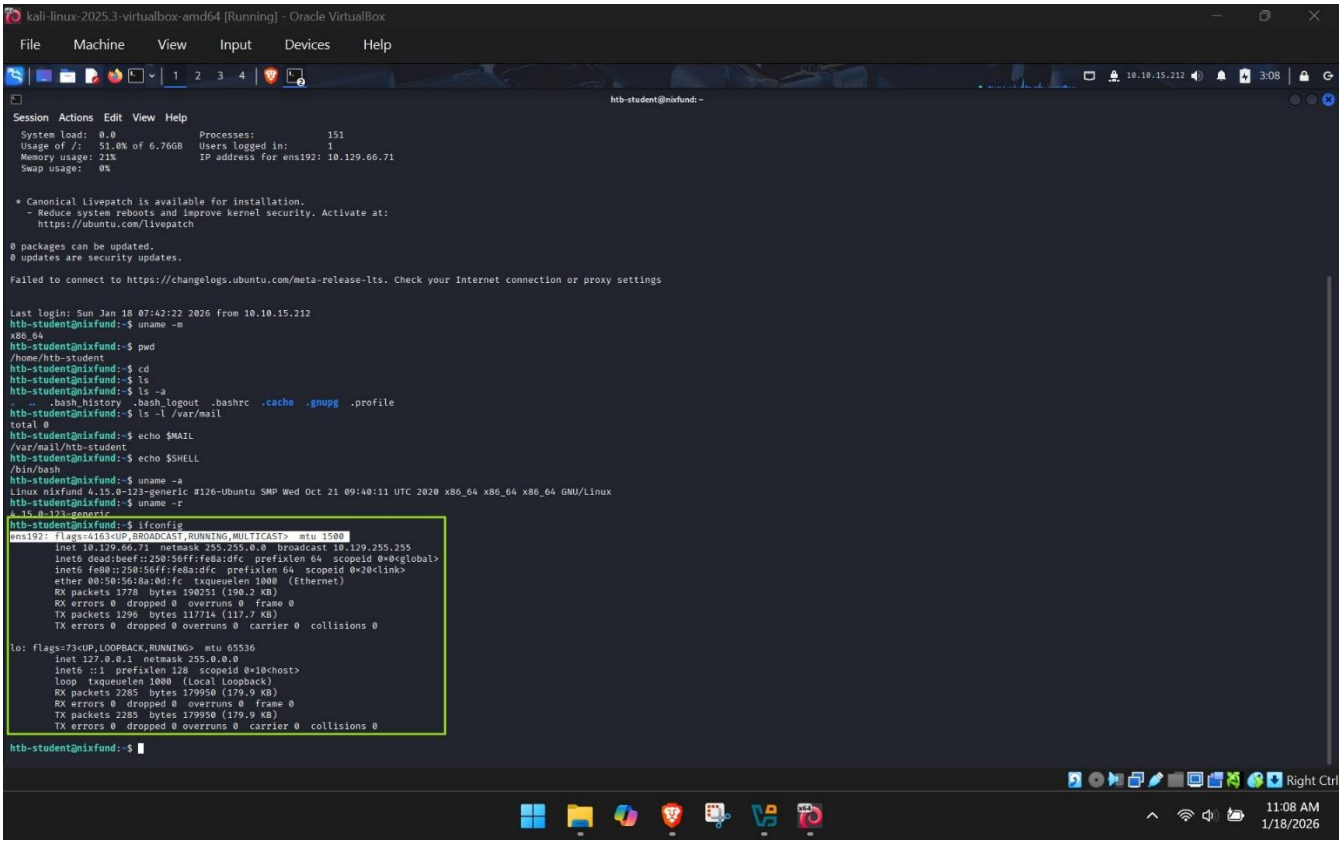


One can run two commands: uname -a or to get the specified kernel release immediately, I ran uname -r.

6. What is the name of the network interface that MTU is set to 1500?

   Answer: ens192

   This is shown in the image below:



The command I ran was ifconfig, which gave me the name of the network interface.

Navigation

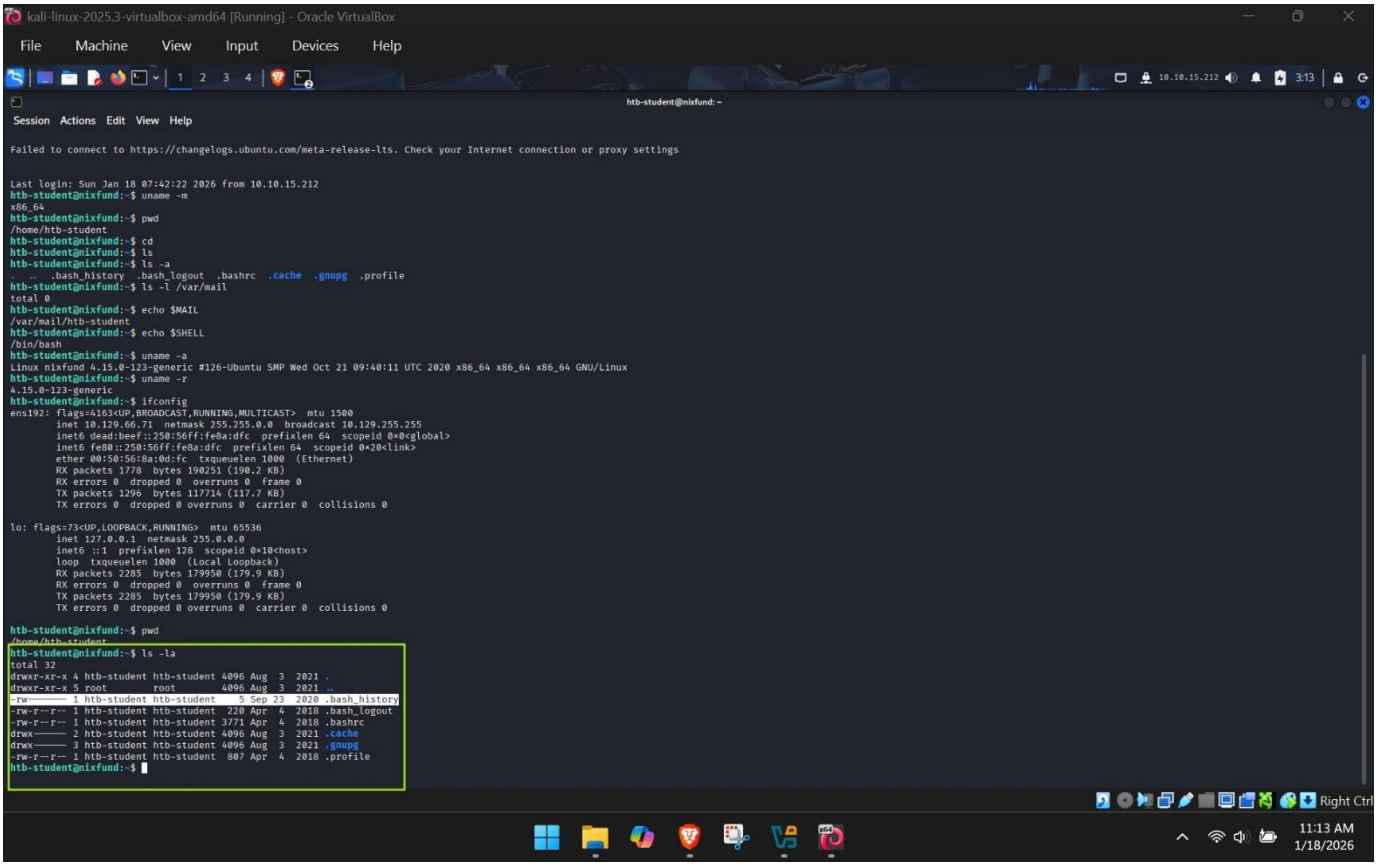
I have learned that navigation is very important, and it is similar to using a mouse as a regular Windows user. I discovered that by navigating, I can get to specific directories and files that I want to work with and move within the system. I also learned different commands and tools for getting information about these files or folders, and I found out that I could use advanced options to alter that output in a way that suits my needs.

Questions

1. What is the name of the hidden "history" file in the htb-user's home directory?

   Answer: .bash_history

   This is shown in the image below:

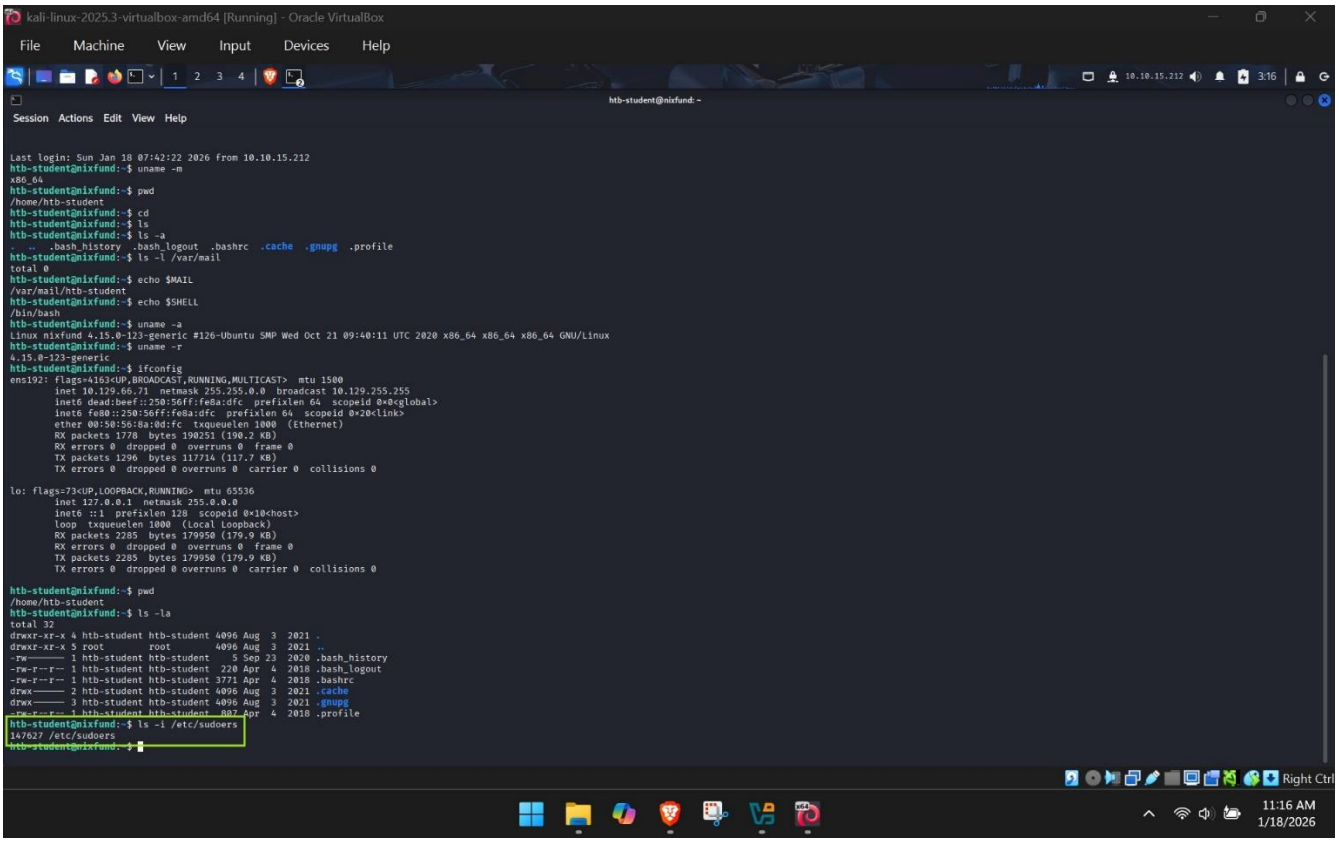


I ran the command ls -la, which its output was hidden directories and files. .bash_history indeed was found out to be the hidden history file

2. What is the index number of the "sudoers" file in the "/etc" directory?

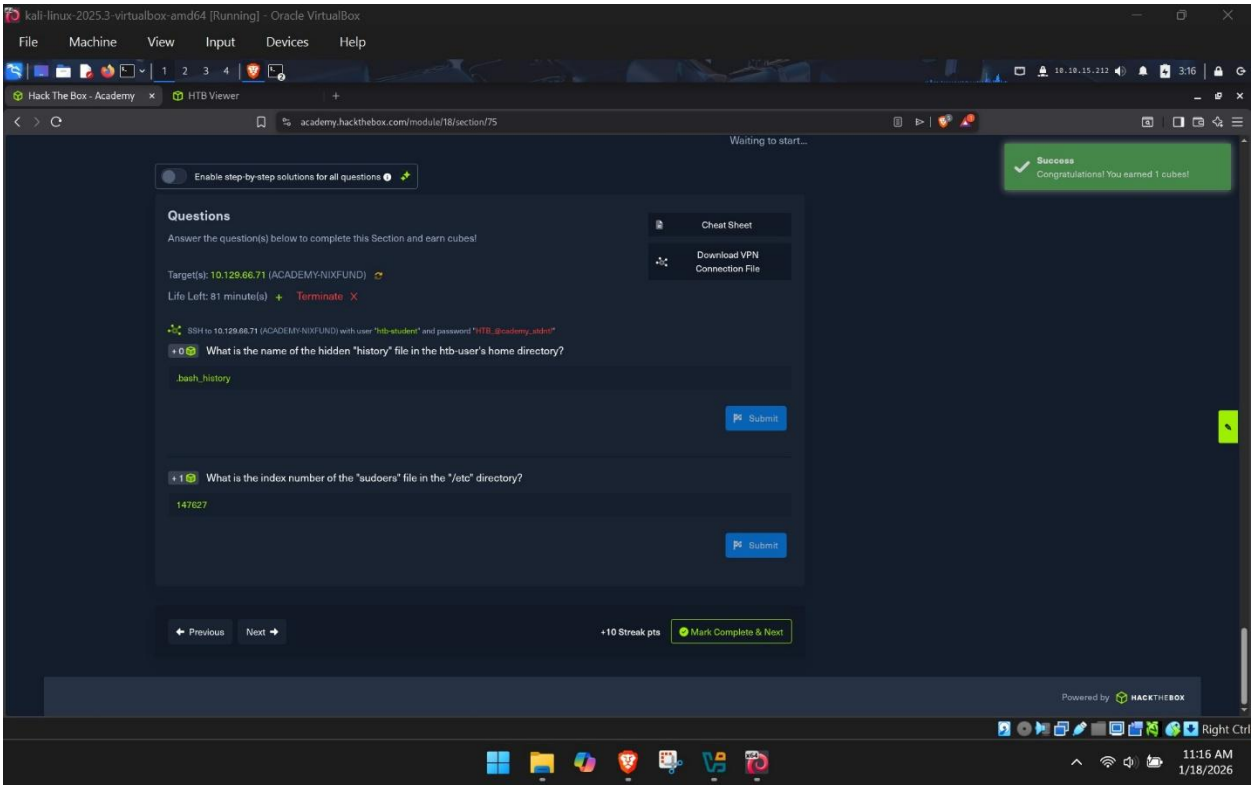
   Answer: 147627

   This is shown below:



I ran the command ls -l /etc/sudoers which gave me the index number.

The proof of questions done is shown below:

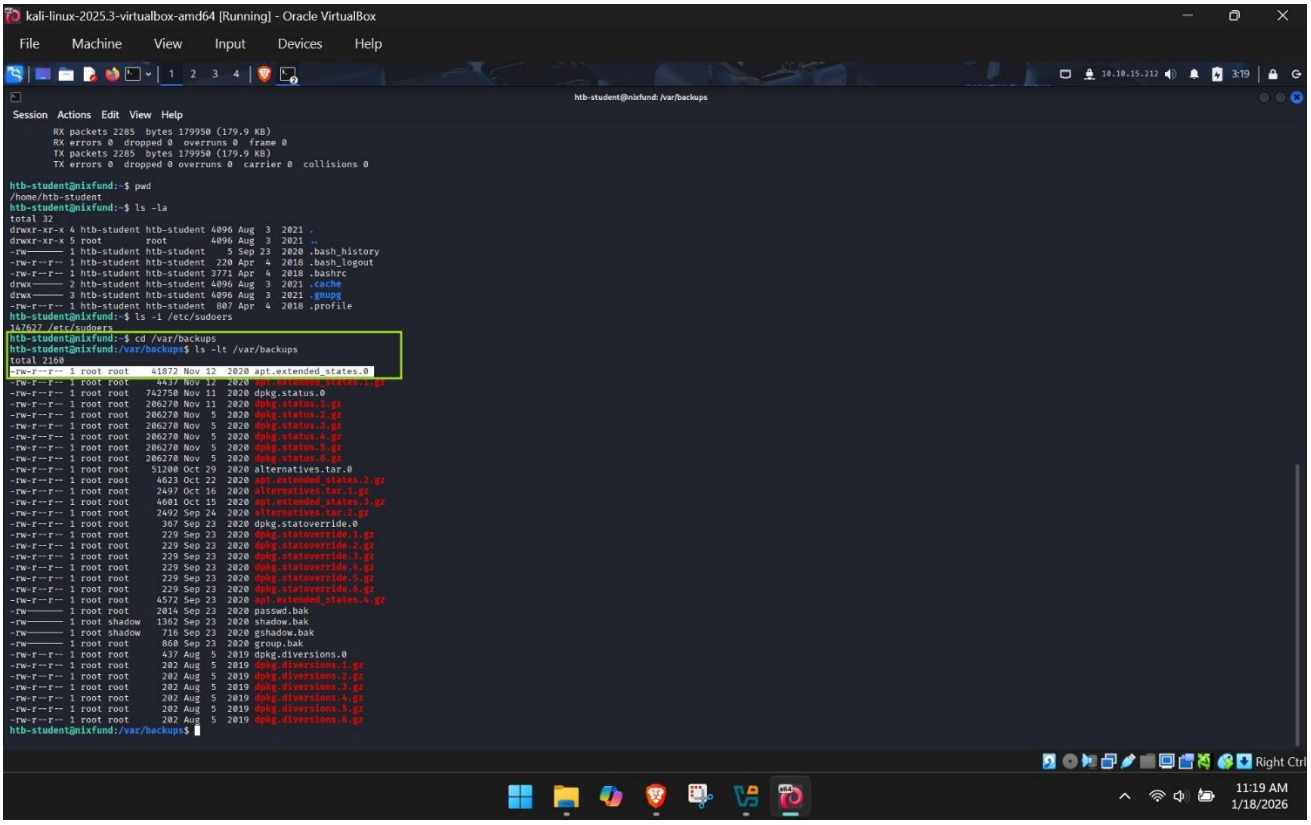


Working with Files and Directories

I realized that the biggest difference between file management in Linux and Windows lies in the way I access and manage files. I noticed that even though I would usually rely on graphical tools such as Explorer in Windows to find and change files, the Linux command line offers a strong option. I have also found that using the command line allows me to access and change files directly, which is a more powerful way of managing files.

Questions

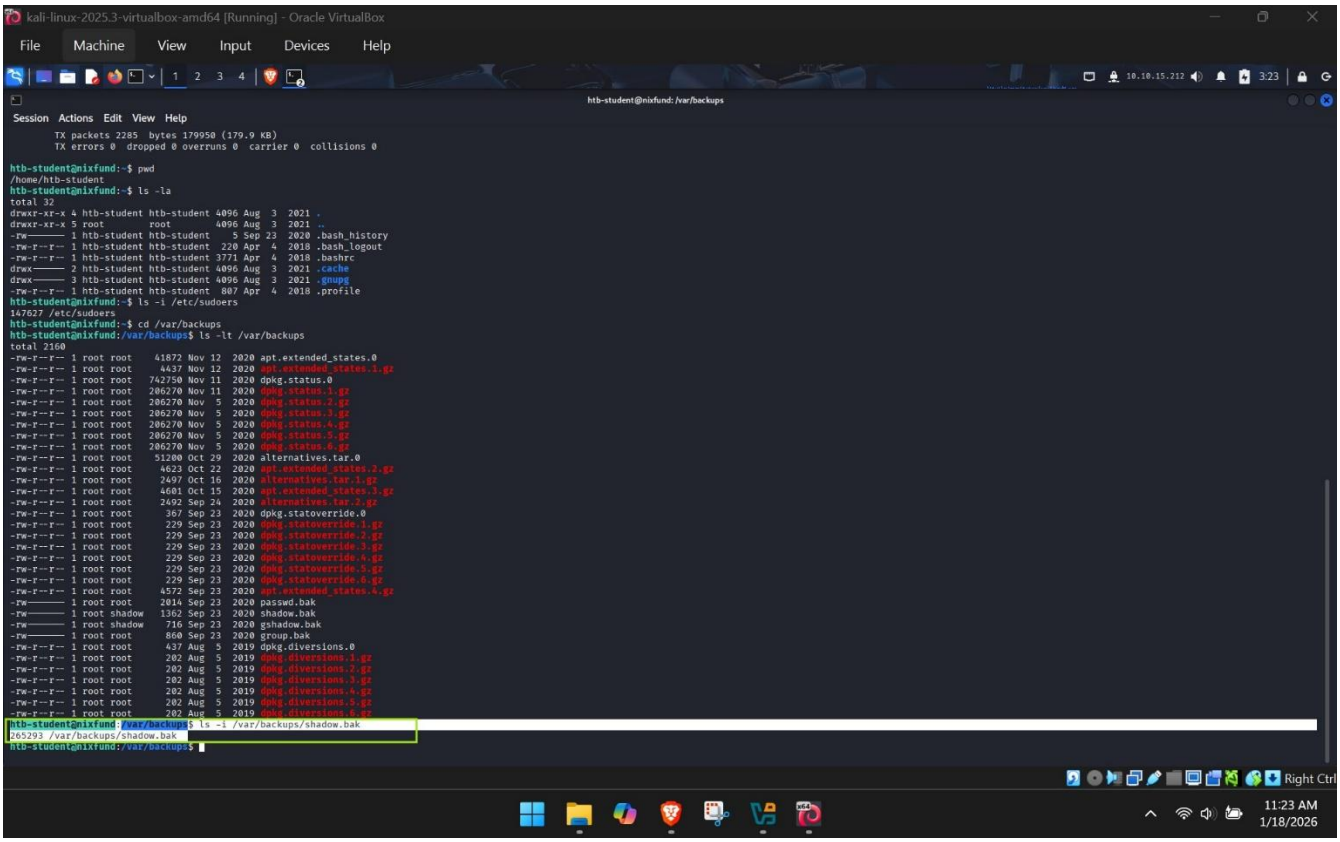
1. What is the name of the last modified file in the "/var/backups" directory?
   Answer: apt.extended_states.0



I used the ls -lt /var/backups and the last modified file is the one on top.

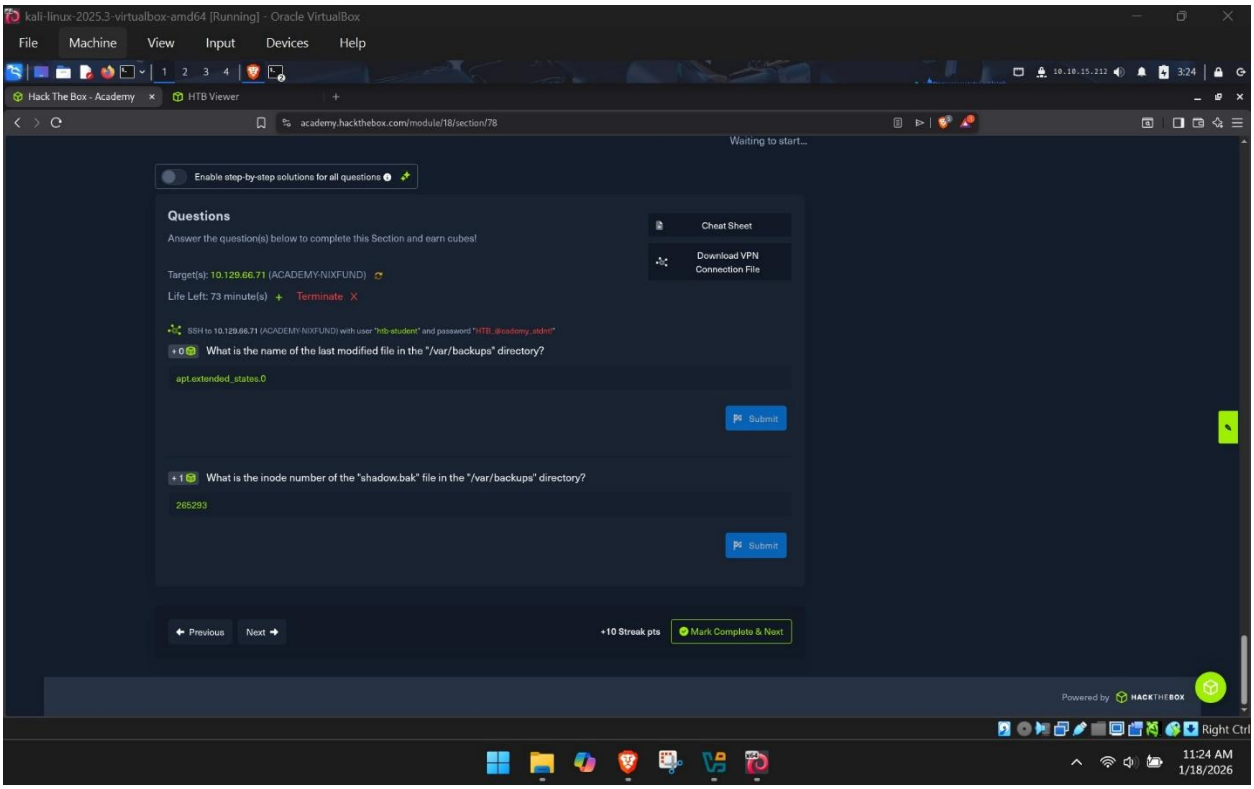
2. What is the inode number of the "shadow.bak" file in the "/var/backups" directory?

   Answer: 265293



I used the command ***ls -i /var/backups/shadow.bak*** which provided the answer.

The proof of questions answered is shown below:



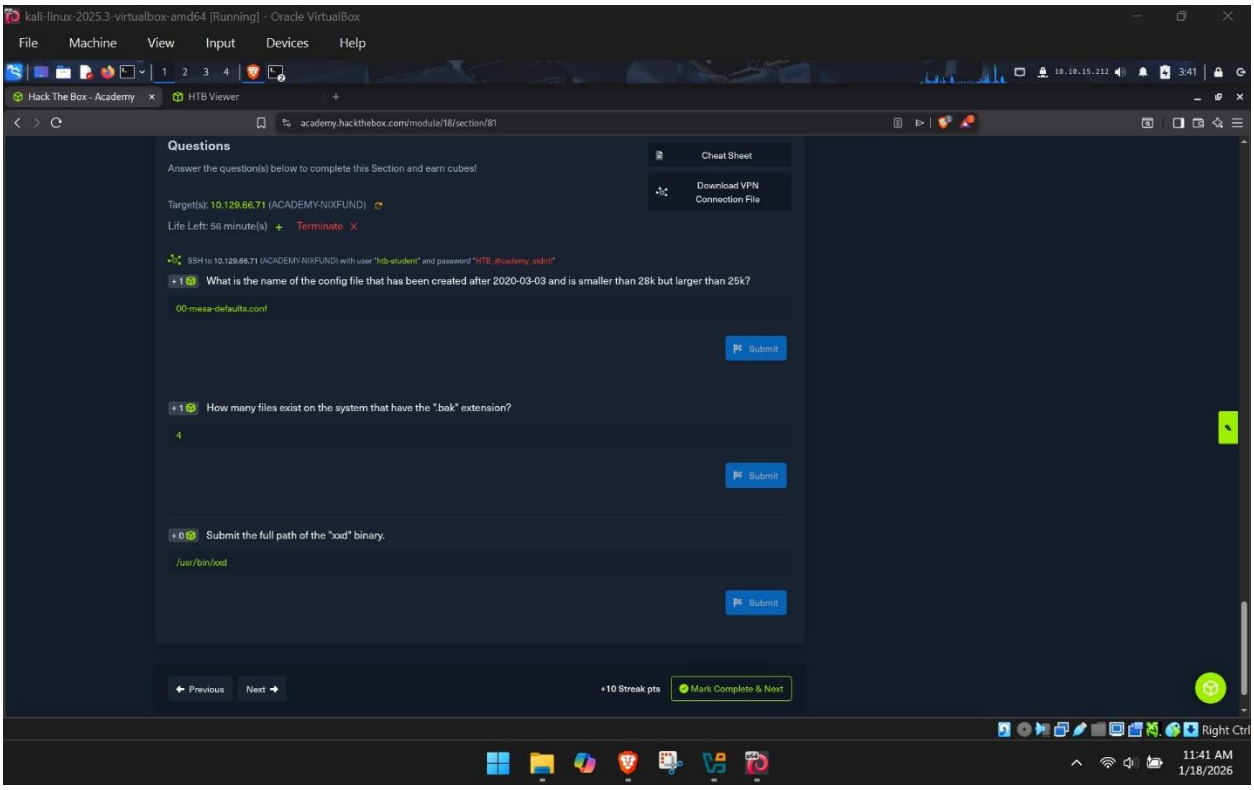
Editing Files

I learned that the most common text editors include vi and vim. Nano is also another text editor that is still widely used.

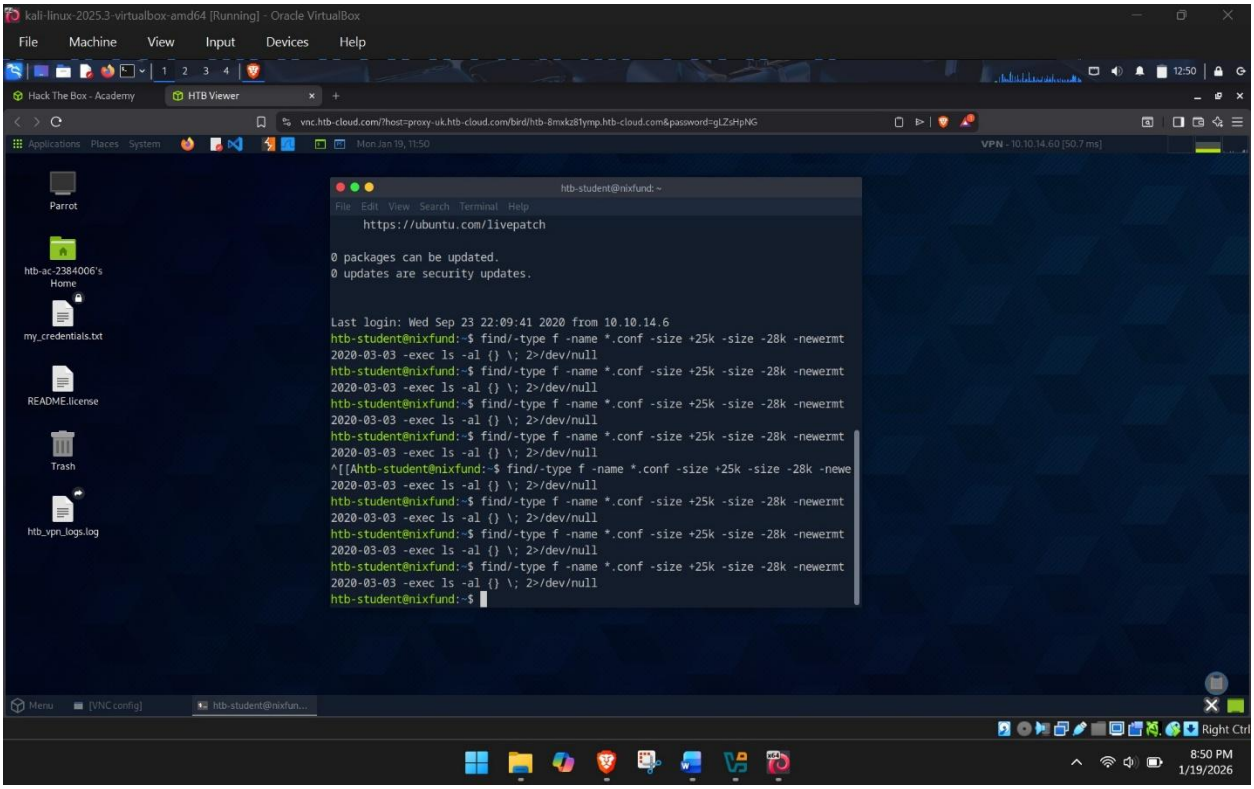
Finding Files and Directories

The first thing that I learned is that it is very important for me to always know exactly where the files and the folders I need are located. I came to the realization that when the time comes that I have been granted access to a Linux system, I will have to locate the configuration files, user-created scripts, and other data that are of utmost importance. Another point that I learned was that it is unnecessary for me to go through all the directories manually or check the timestamps of the files one by one because there are already efficient means of identifying modified items' last times.

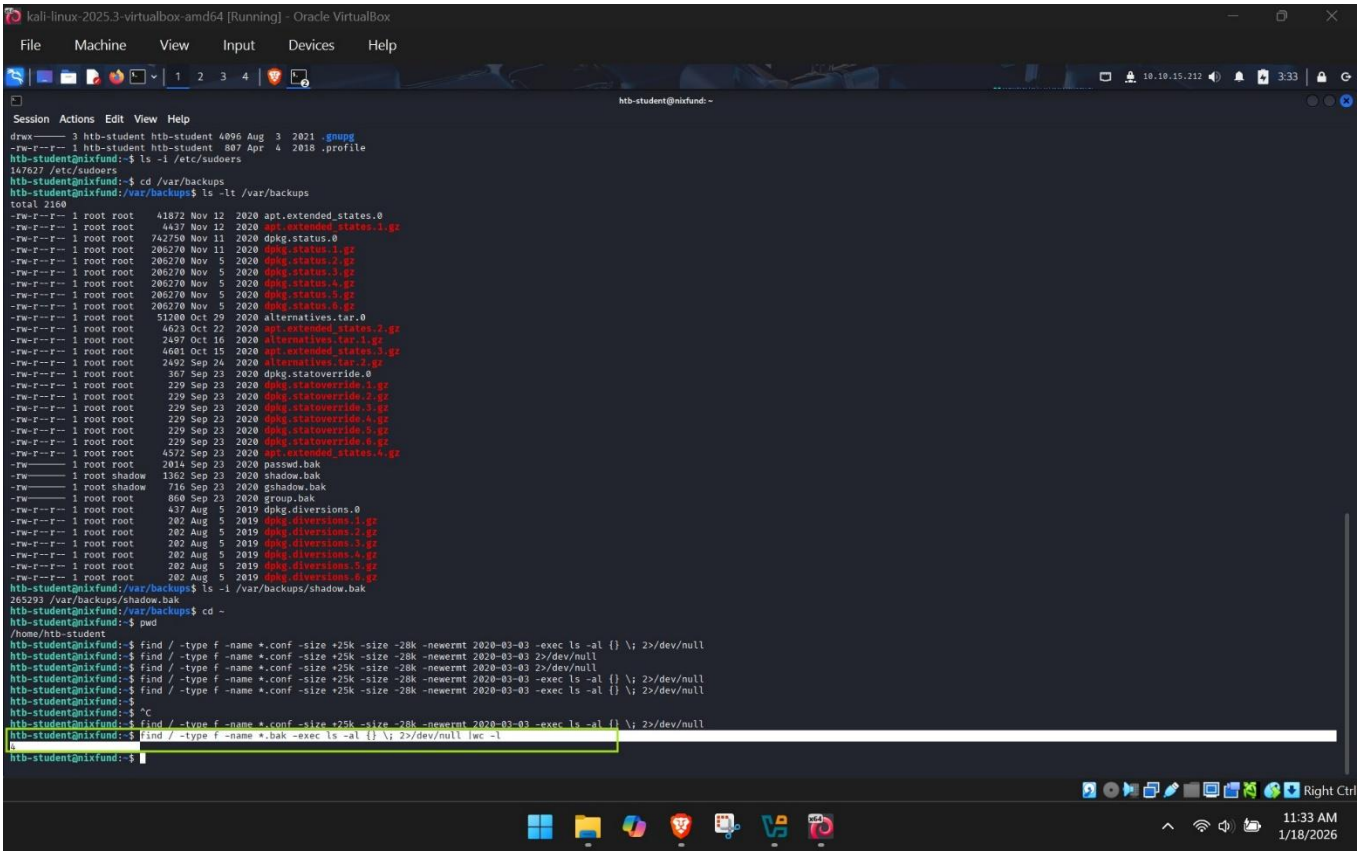
Questions



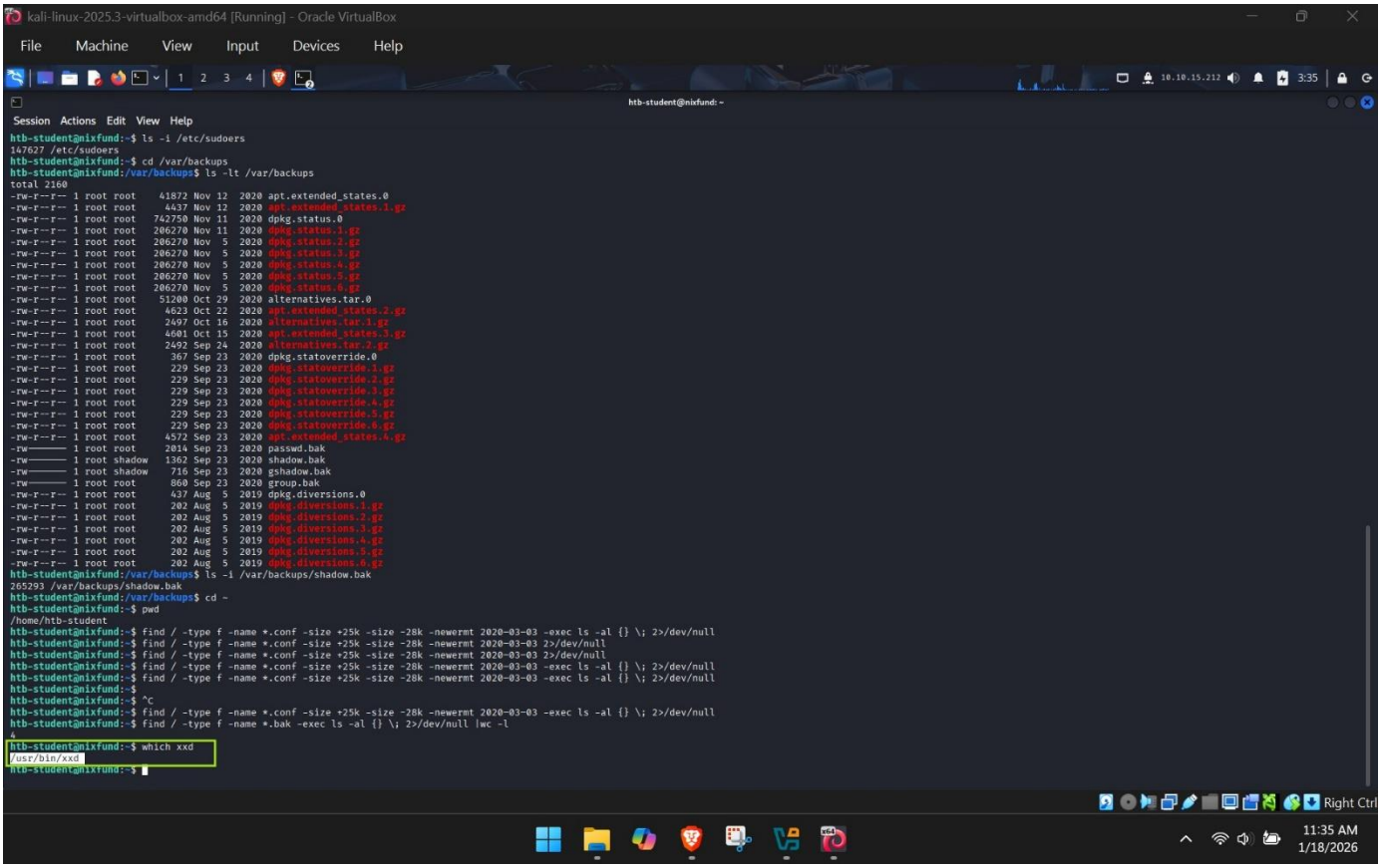
Question 1: The command I used was ***find / -type f -name *.conf -size +25k -size -28k -newermt 2020–03–03 -exec ls -al {} \; 2>/dev/null'*** which did not really give me any output on my end, despite using the pawnbox.



Question 2: The command I ran was ***find / -type f -name *.bak -exec ls -al {} \; 2>/dev/null |wc -l*** which gave me the output as shown below:



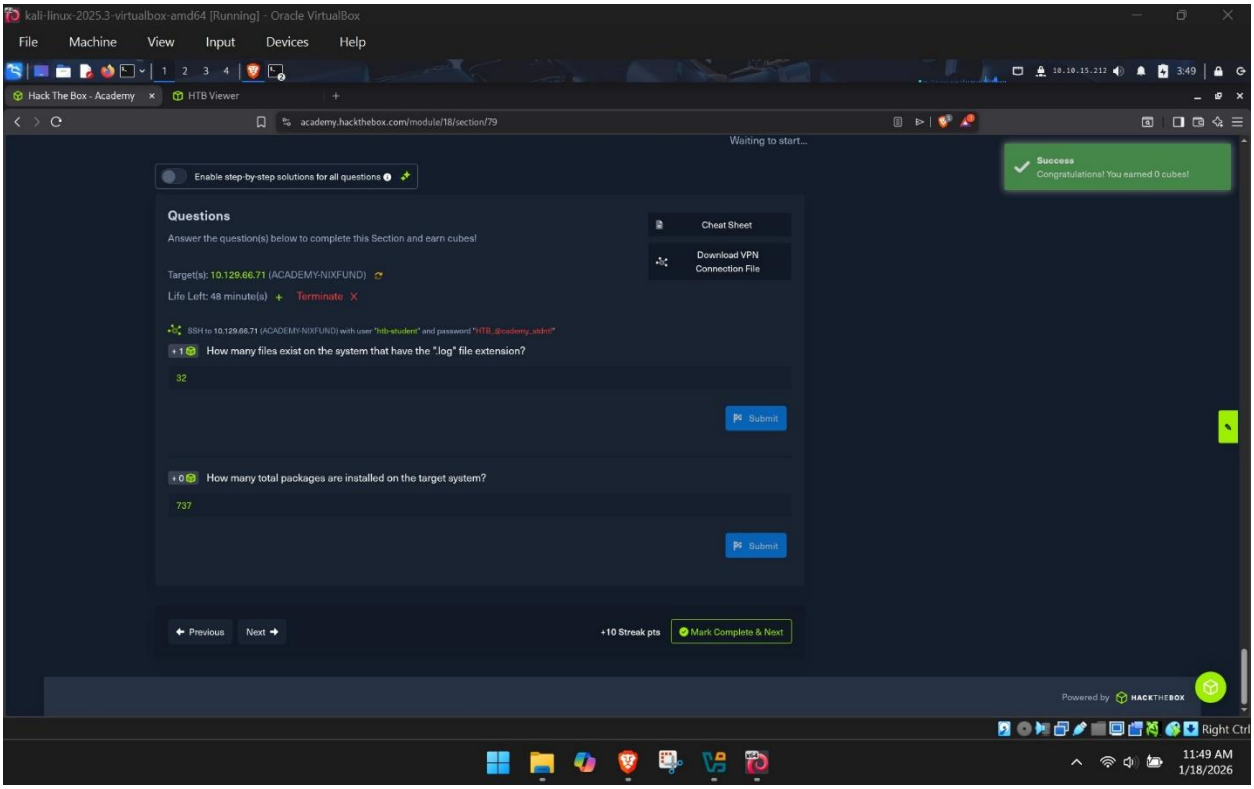
Question 3: The command I ran was which xxd which gave me the output as shown below:



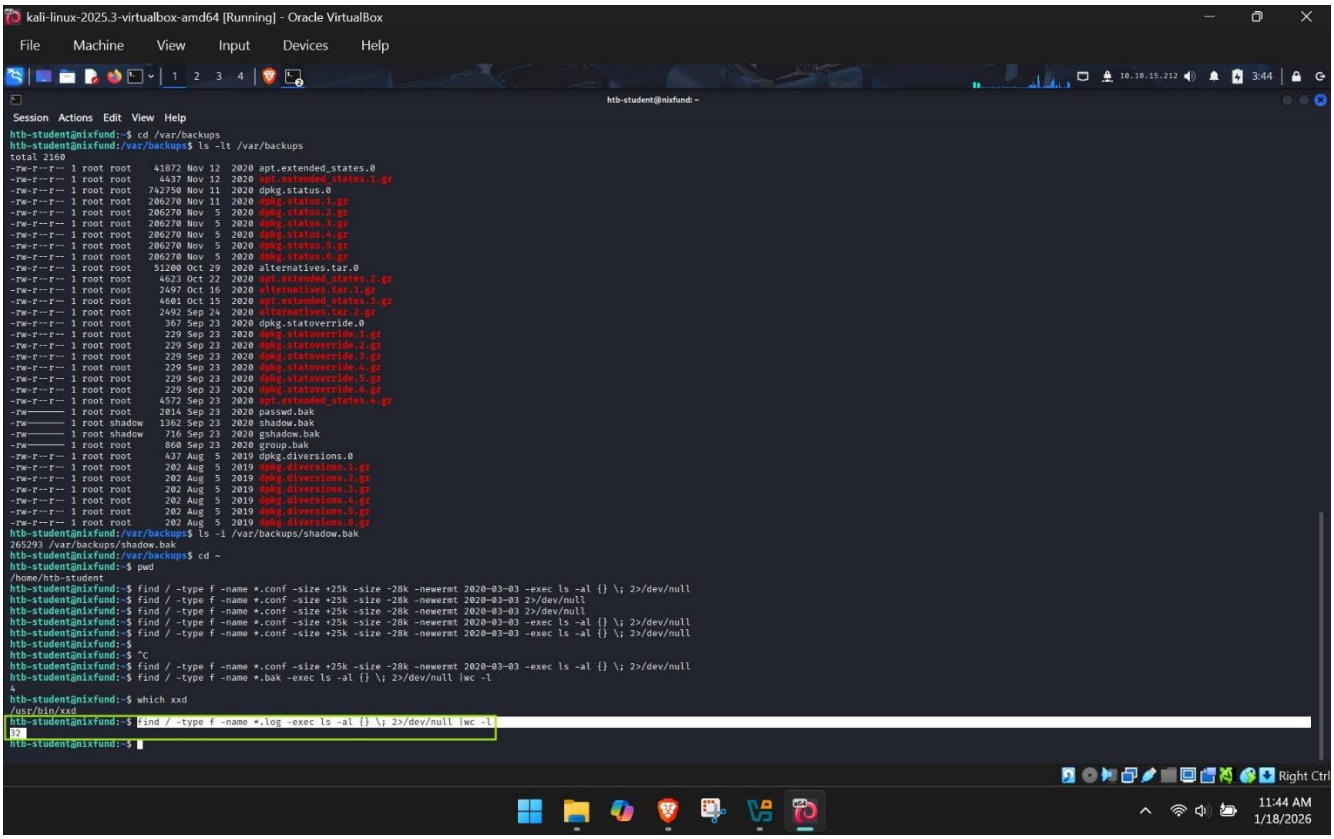
File Descriptors and Redirections

I understood that a file descriptor (FD) in Unix and Linux systems is a reference which is under the control of the kernel and is used for the purpose of I/O operations. I also found out that it acts like a unique id for the I/O resources, e.g., open files, sockets, and so on, and that it is the same as what is usually termed a file handle in Windows. Moreover, I realized that a file descriptor is basically the system's way of monitoring all the active connections, like when I am reading from or writing to a file.

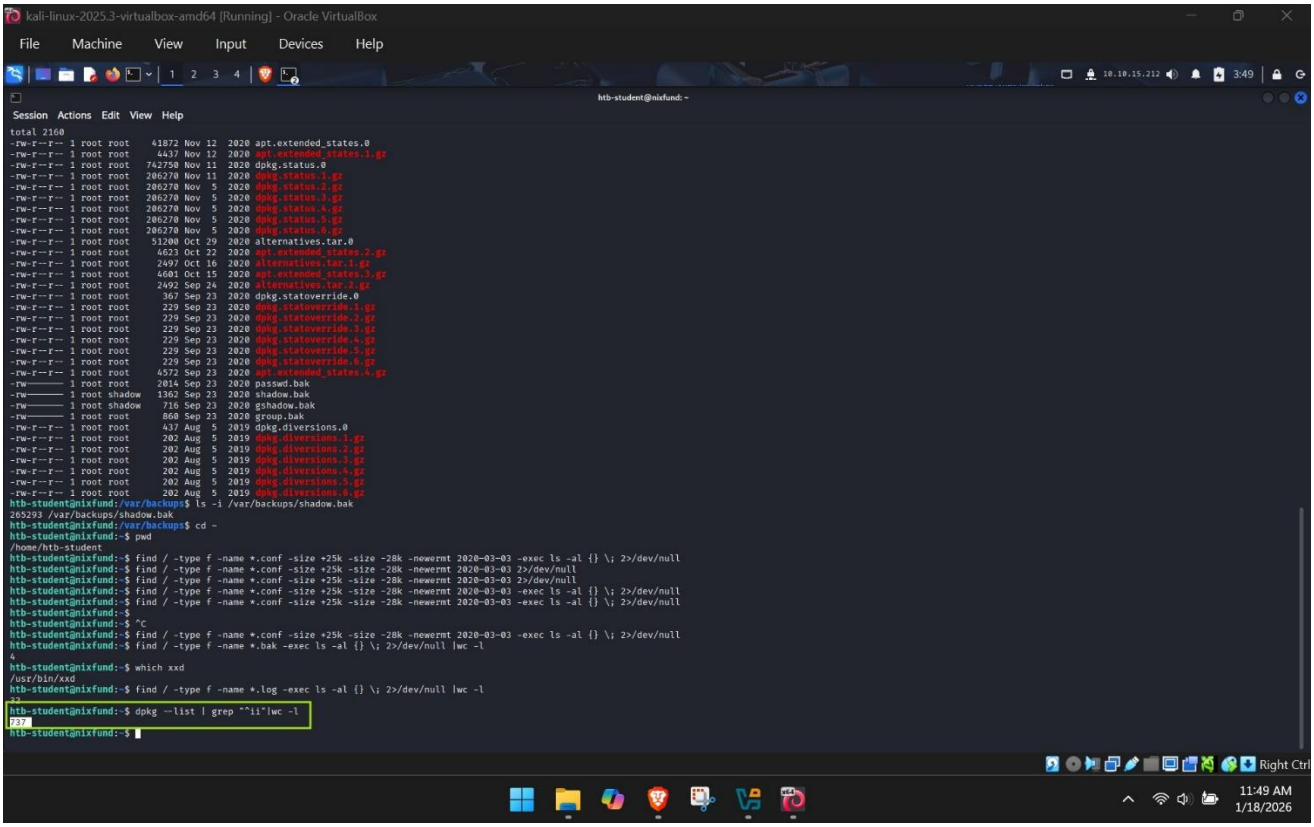
Questions



For question 1, I ran the command ***find / -type f -name *.log -exec ls -al {} \; 2>/dev/null |wc -l*** and the output is shown below:



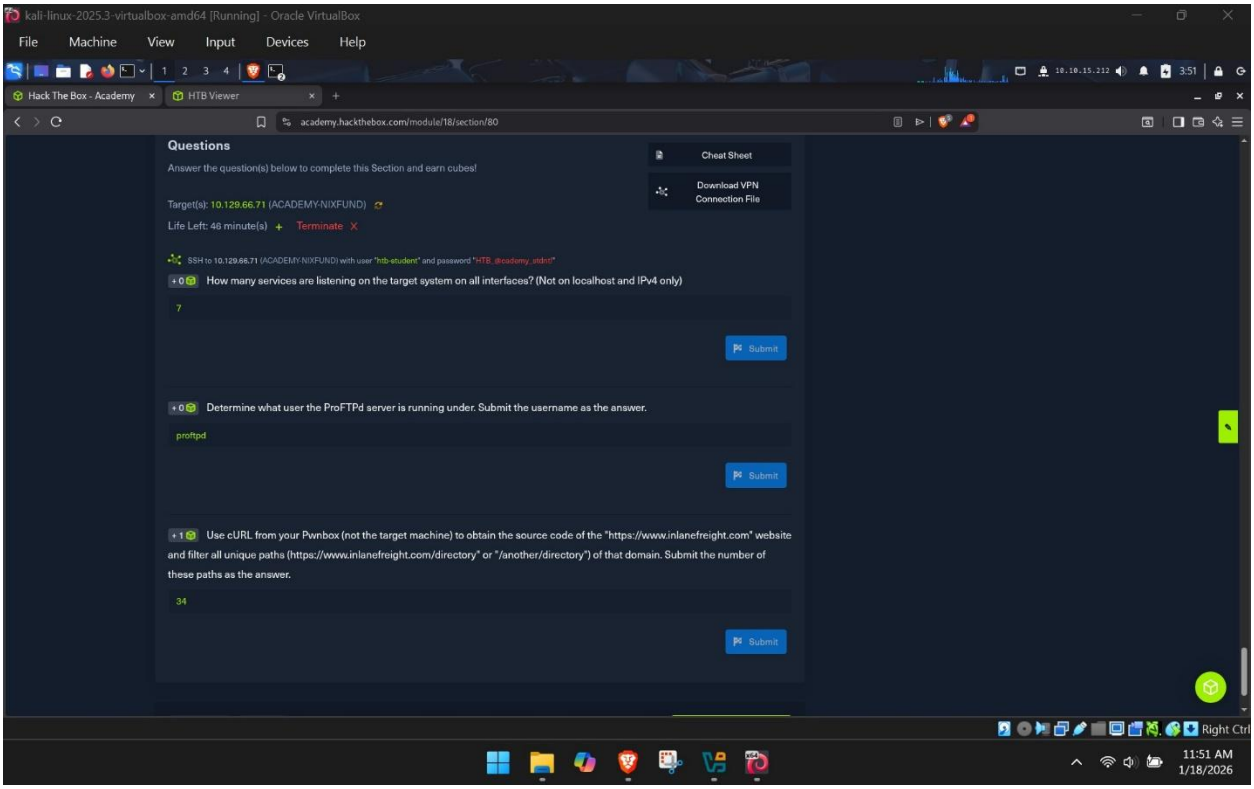
For question 2, I ran the command ***dpkg –list | grep "ii" | wc -l*** and the output is shown in the image below:



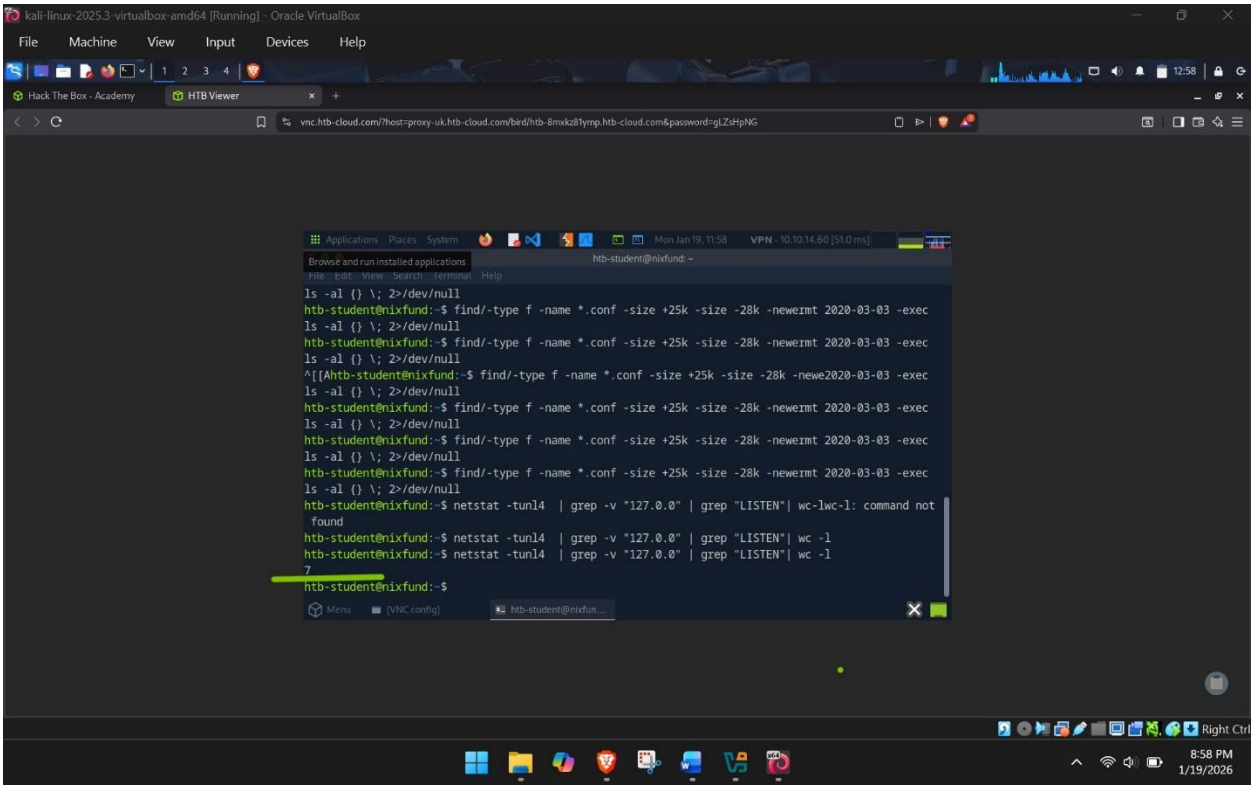
Filter Contents

I learned on how one can read file contents without the use of a text editor; with two powerful tools: more and less.

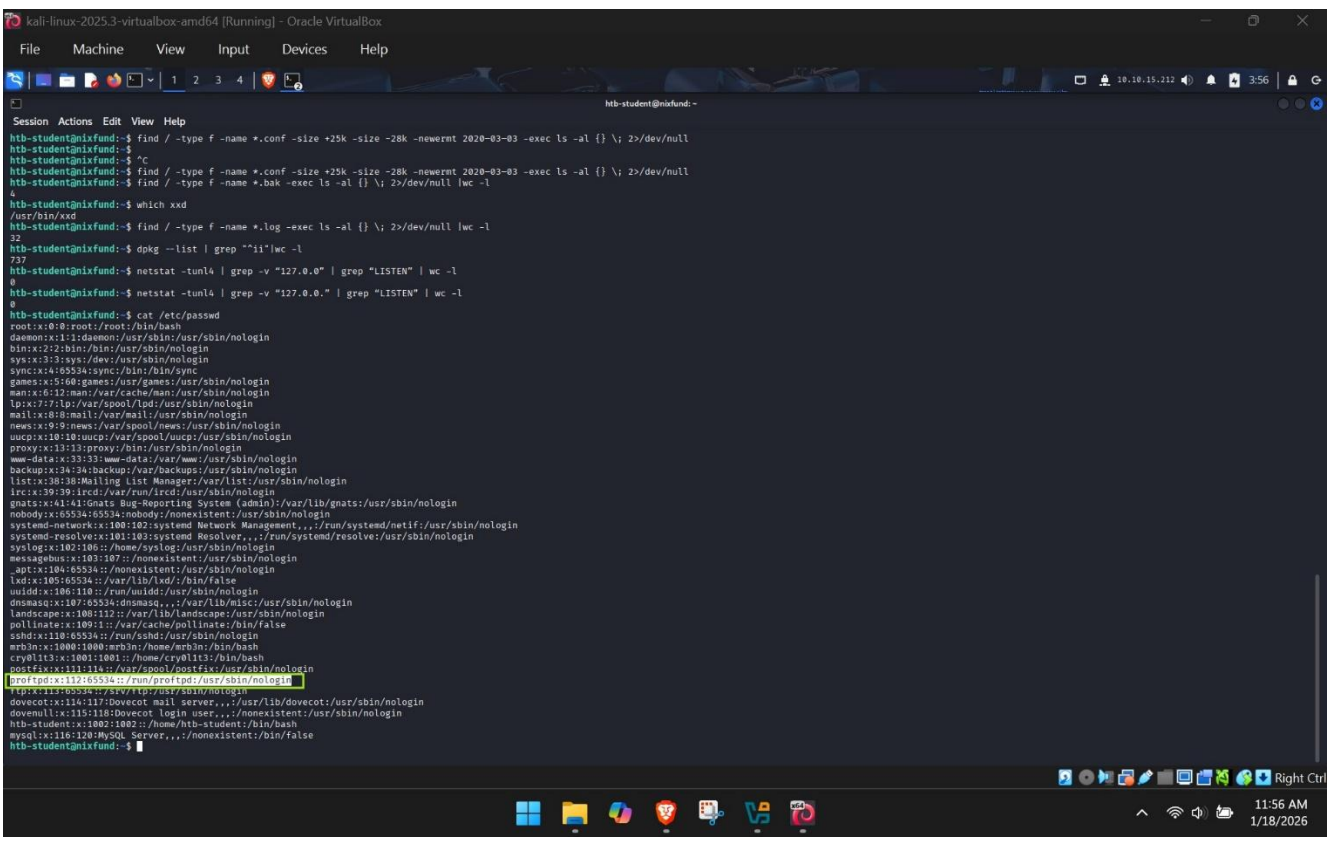
Questions



For question 1, I ran the command ***netstat -tunl4 | grep -v "127.0.0" | grep "LISTEN" | wc -l,*** of which the output is shown below:



For question 2, I used cat /etc/passwd in order to get the proFTPd server is shown below:



For question 3, I was unable to find the right answer, since my output was 39, which was the wrong answer.

Regular Expressions

I came to learn that regular expressions (RegEx) are nothing but the art of developing precise blueprints for finding patterns in text or files. They give one the power to accurately and precisely perform data identification, swapping, and modification with an astonishing degree of precision.

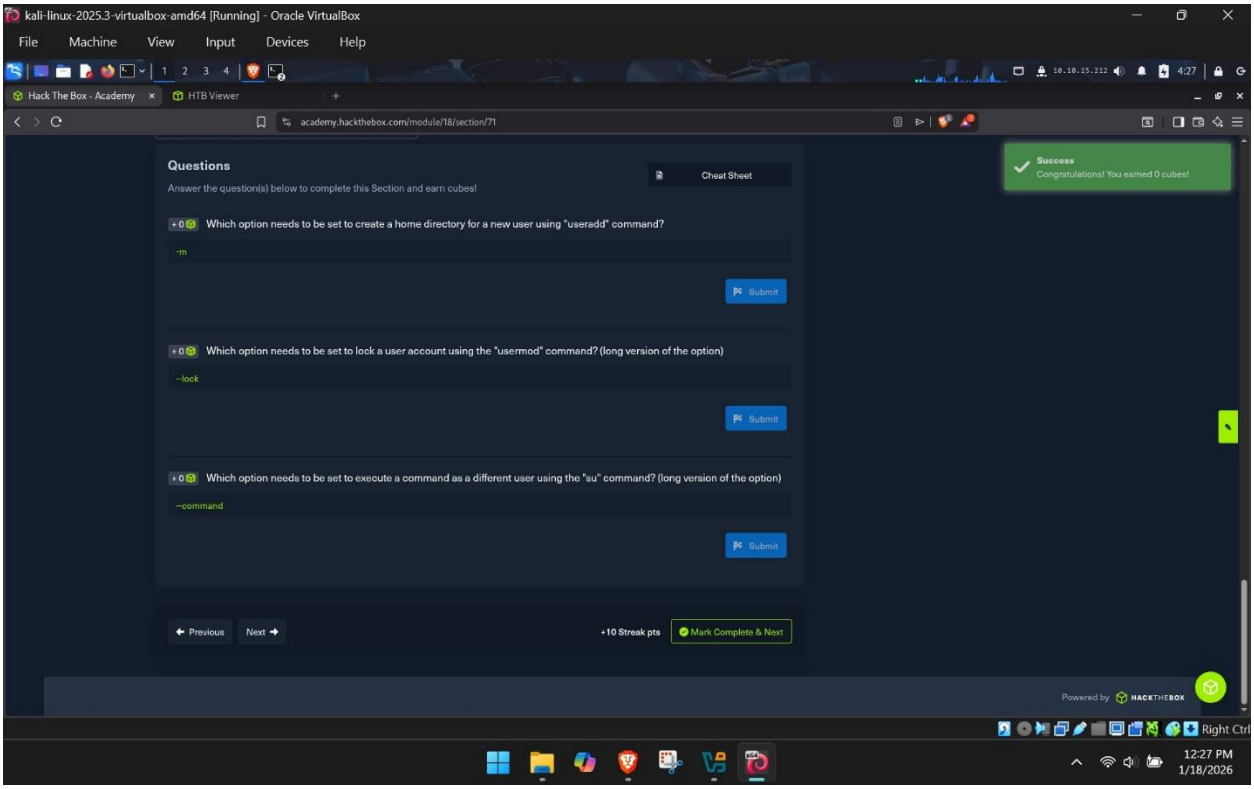
Permission Management

I came to know that Linux permissions operate like keys that control the access of different users to the files and directories. I see these permissions as being given to single users and groups of users, much like access is granted to different people and groups in a company. I also found out that one user can belong to several groups and that this membership gives the user additional rights to carry out certain operations on files and folders.

User Management

I came to know that Linux system administration entails the efficient management of users which is a major aspect. I have as an administrator often to keep access rights properly, create new user accounts or put the existing ones into specific groups. Also, I realized that changing the user while running the command is a skill that is useful very often for dealing with tasks requiring different levels of privilege.

Since this was a theoretical based section, the questions as well as their answers are shown below:



Package Management

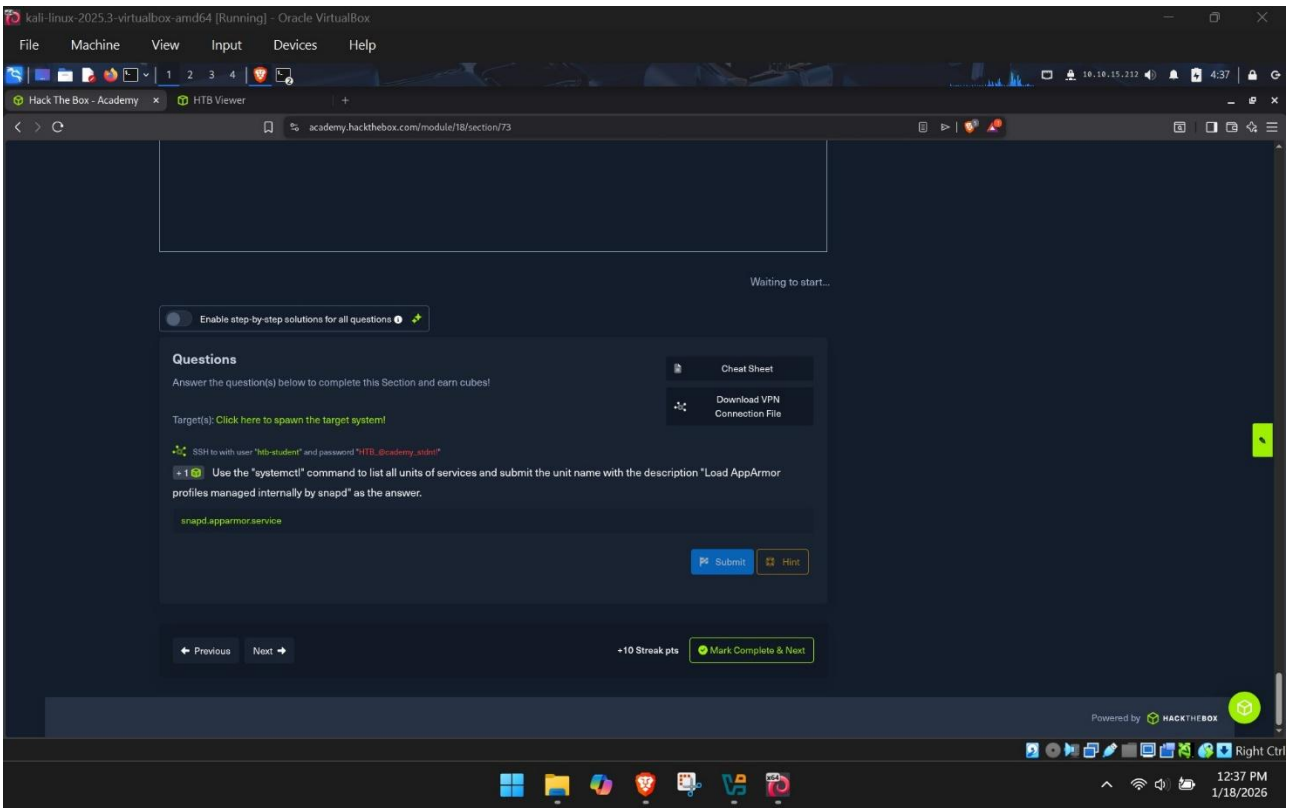
I have come to know that it does not matter if I am a system administrator, taking care of my private Linux devices at home, or setting up and maintaining my favorite penetration testing distribution, I should really know well the Linux package managers that are at my disposal. It is on my part to have a thorough knowledge of the various ways to use these tools in order to be able to do software package installation, updating, or uninstalling in an efficient manner.

Service and Process Management

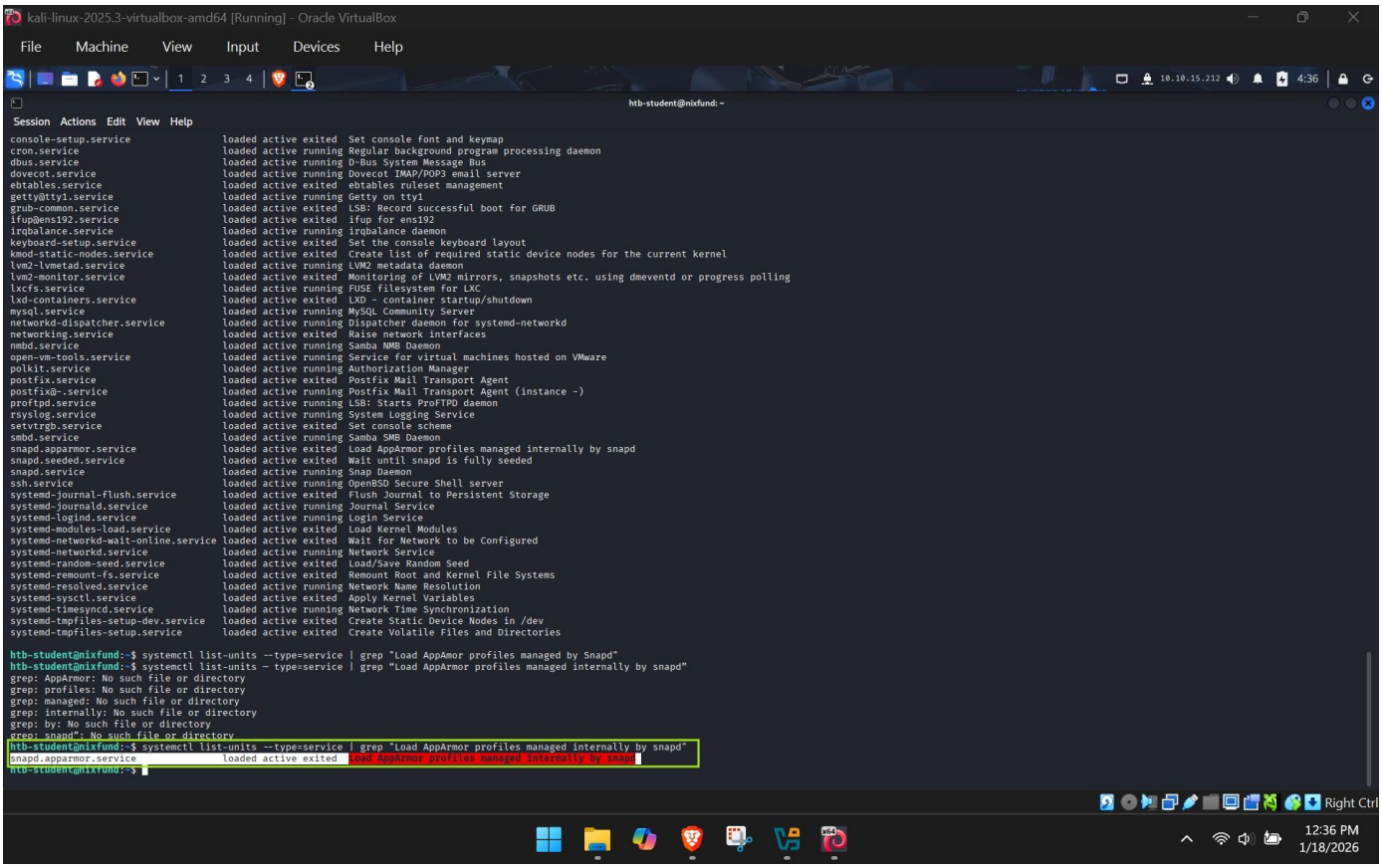
I learned that services, which are often called daemons, are of utmost importance to Linux and they work silently in the background without any need for me to interact with them directly. The services, in fact, perform very important functions for the system like making it stable and giving

it some more features. Moreover, I got to know that services are typically classified into two major classes: the ones that are part of the system and the ones that are added by the user.

Question



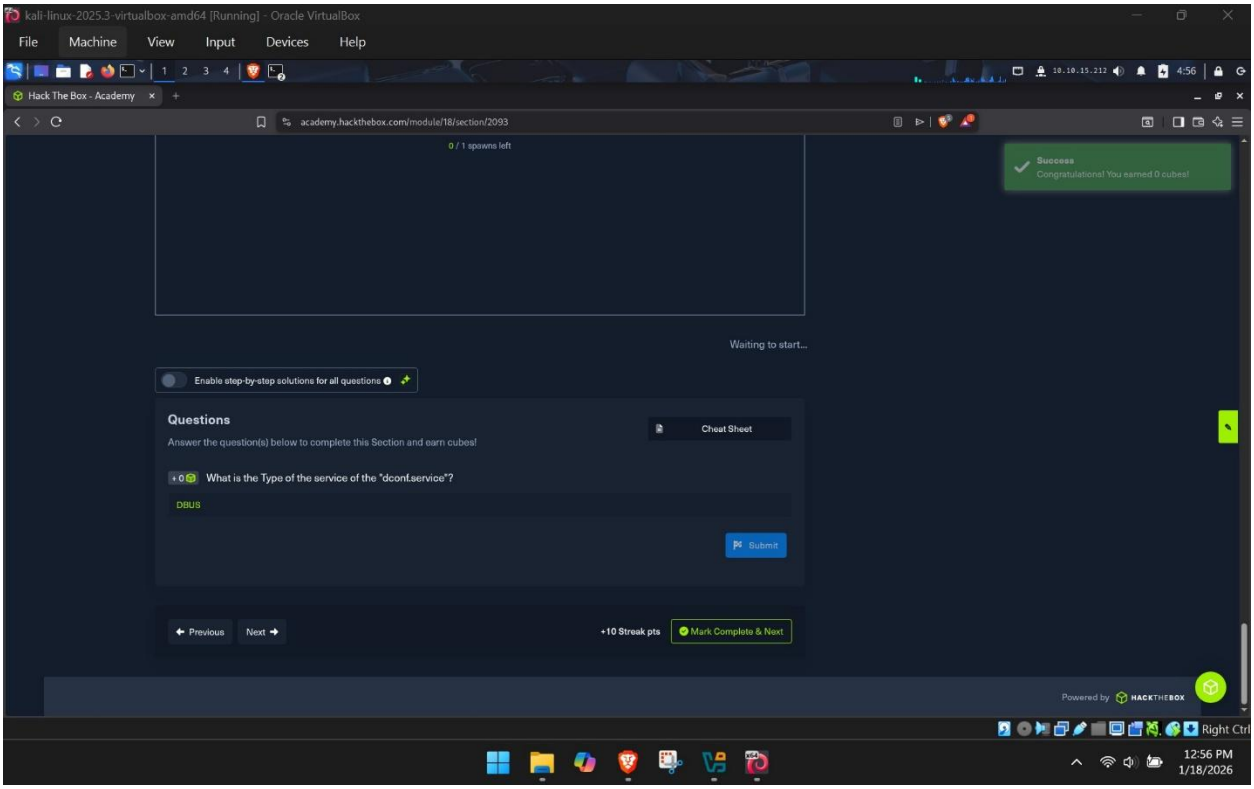
I used the command ***systemctl list-units — type=service | grep "Load AppArmor profiles managed internally by snapd*** in order to get the unit name. This is shown below:



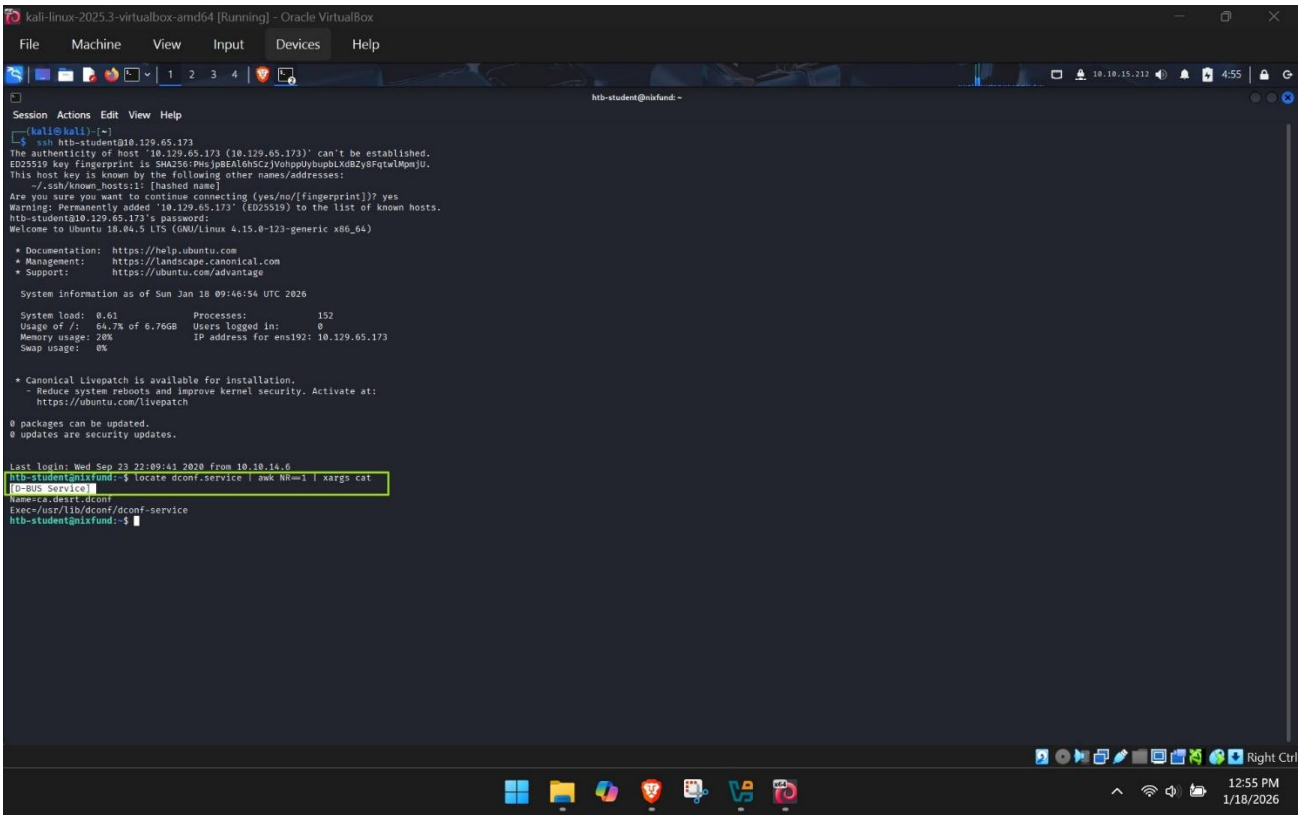
Task Scheduling

The key thing I learned is that Linux task scheduling function is to automate excellent feature that I can execute my processes at exact times or even at regular intervals. I came to the conclusion that this feature is indispensable for both the users and the administrators as it completely eliminates the need of starting the tasks manually.

Question



I ran the command ***locate dconf.service | awk NR==1 | xargs cat*** to get the type of service.



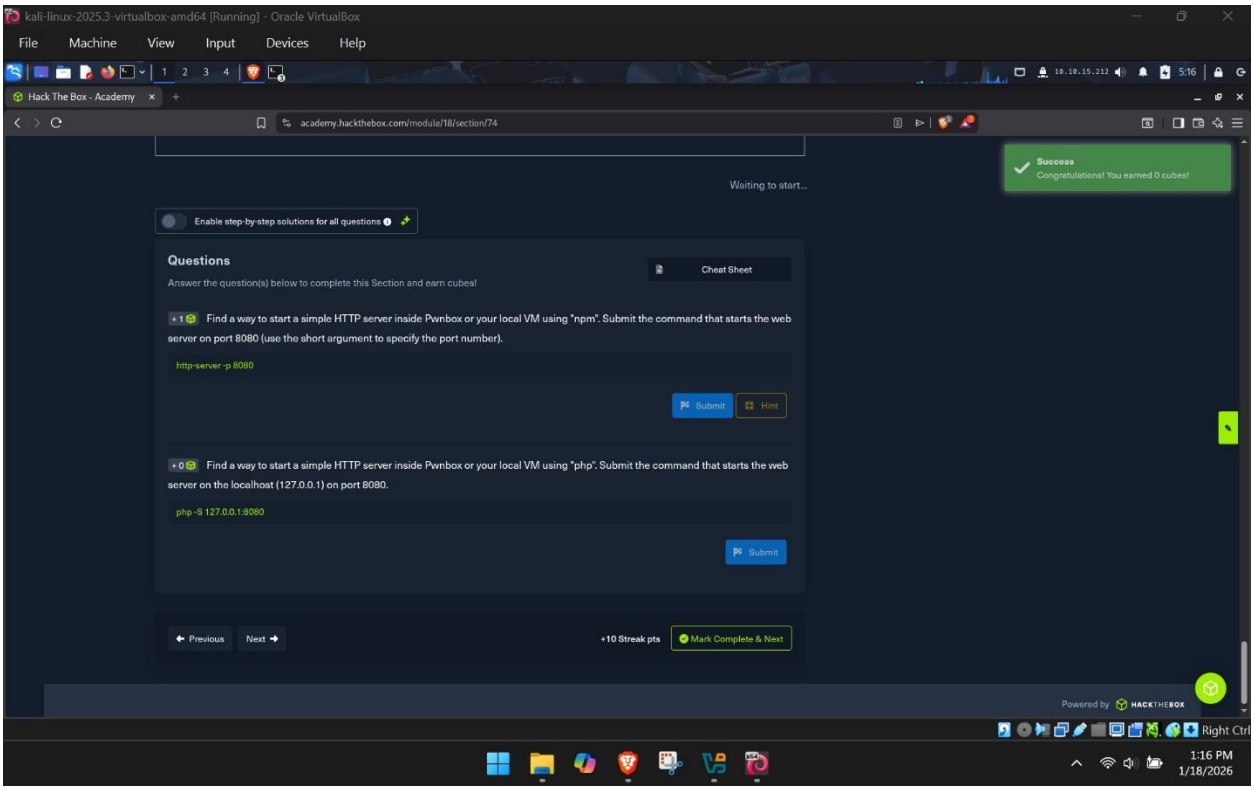
Network Services

I discovered that controlling various network services is one of the main parts of Linux usage. I realized that being skilled in managing these services is very important, mainly due to the fact that they are designed to carry out particular functions that help in remote work.

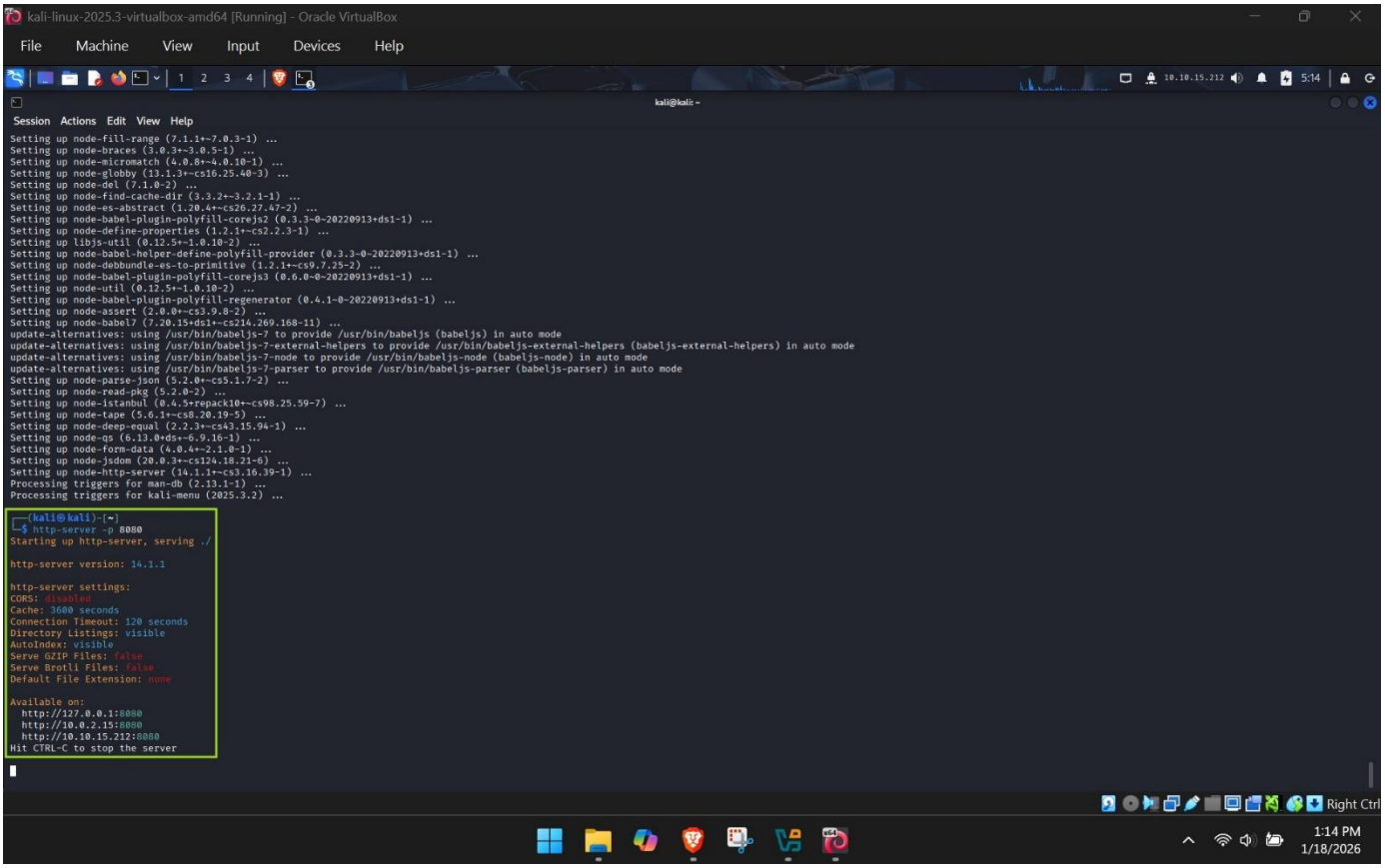
Working with Web Services

Communication between browsers and web servers is another important aspect of web development that I have learned about. I explored the different ways of setting up a web server on Linux OS and found that Nginx, IIS, and Apache are the most popular. Also, it was revealed to be the case that among the various web servers, Apache is the one mostly used in the industry.

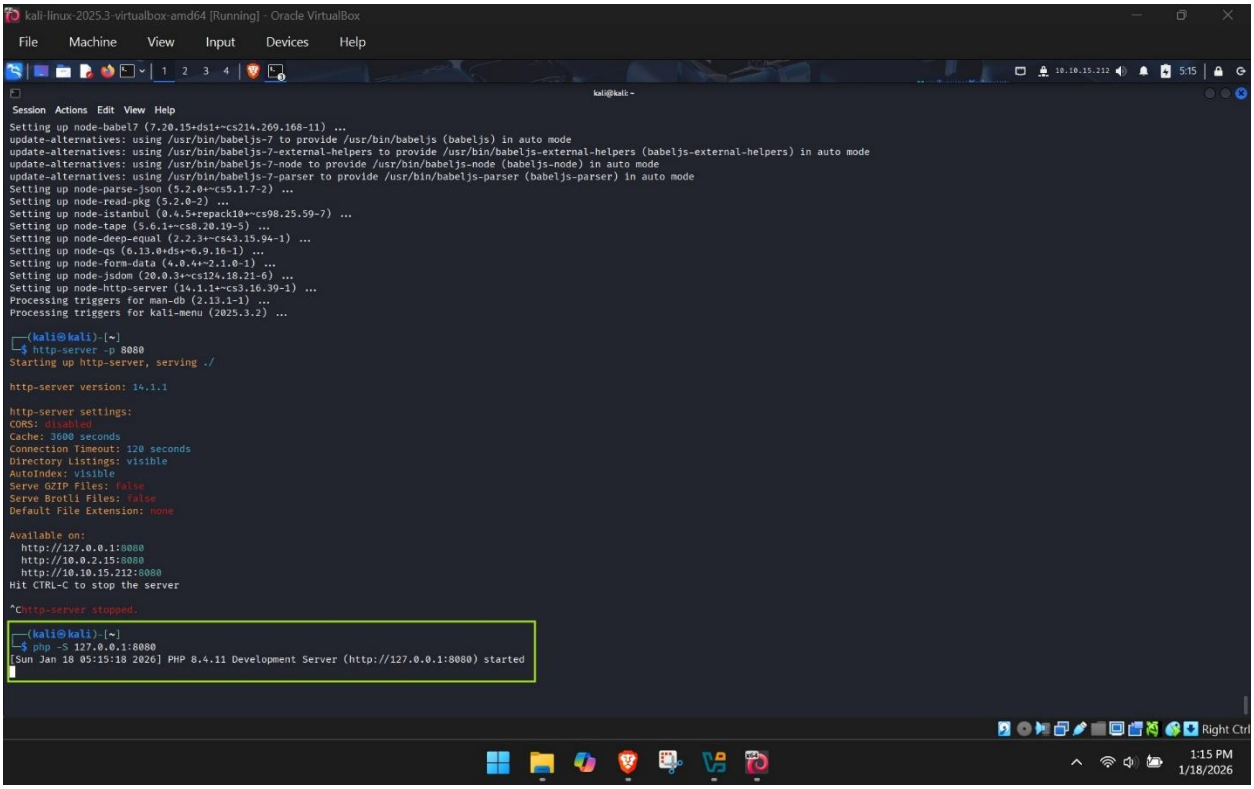
Questions



For the http server, I ran the server by typing http-server -p 8080, which was able to open the server as shown below:



For the php server, I ran the php server by typing php -s followed by the IP address and the port, as shown below. The server was able to open.



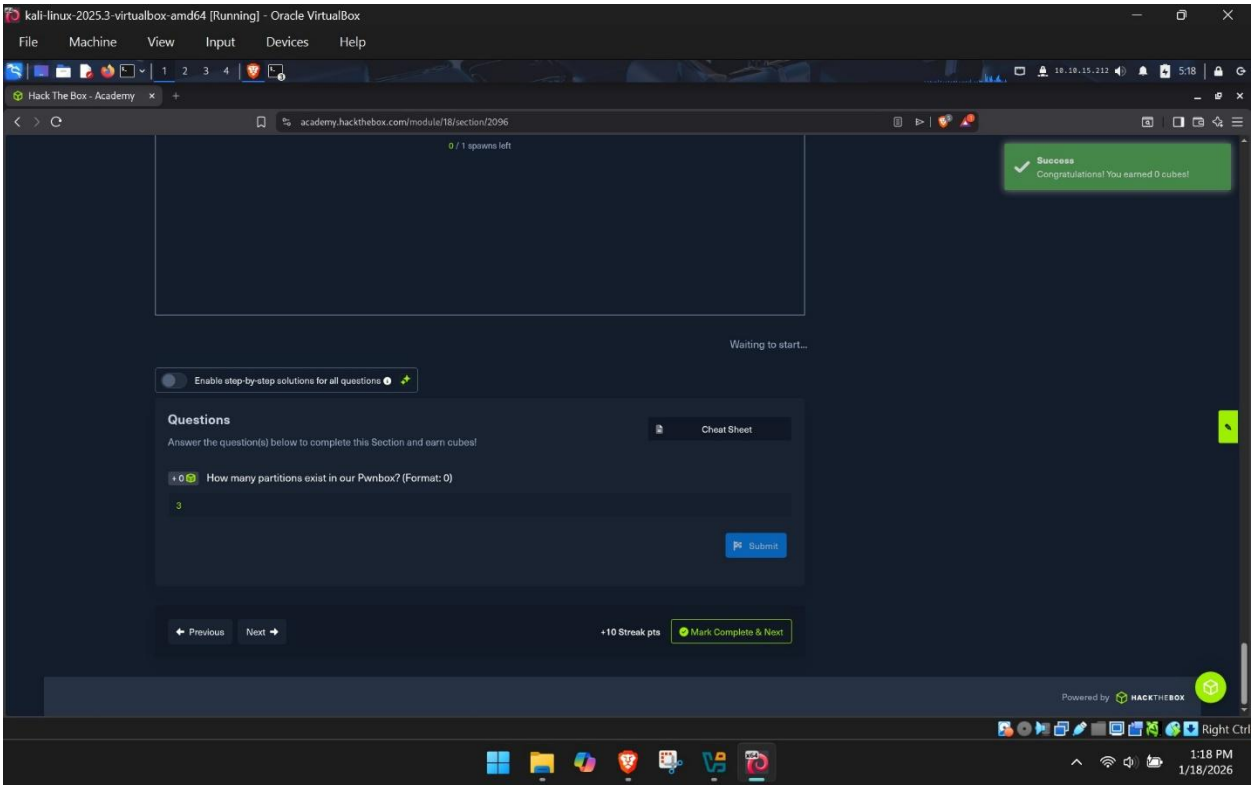
Backup and Restore

I learned that Linux has its own powerful tools for backing up and restoring data such as Rsync or Deja Dup.

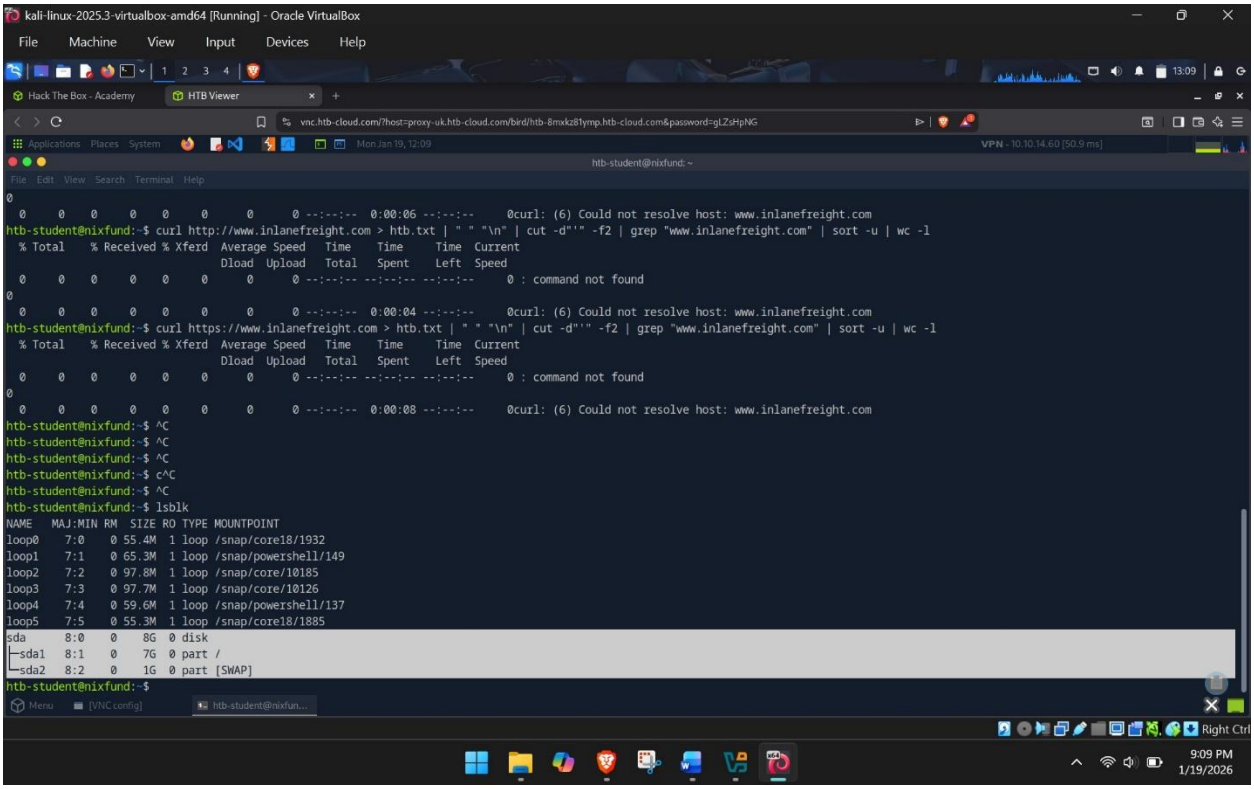
File System Management

File system management on Linux is a critical task that includes organizing, storing, and keeping data safe on disks or any other storage devices, and this is a skill I have learned. Linux turned out to be an extremely flexible operating system that can work with many file systems including ext2, ext3, ext4, XFS, Btrfs, and NTFS. Furthermore, I became aware of unique features of each file system, hence, their suitability for different applications and scenarios.

Question



I used the command lsblk to list all the block devices as well as their partitions. The output displayed one disk, sda, with three partitions.



Containerization

I learned that containerization is a method of packaging and running applications in separate spaces called containers. I found these containers to be light and stable environments, providing that the applications would be consistent wherever I deployed them. Furthermore, I discovered the existence of some technologies such as Docker, Docker Compose, and Linux Containers (LXC) that support this process mainly in Linux environments. I learned that containers are different from virtual machines in that they utilize the host's kernel, which results in much higher efficiency and lower resource consumption. Lastly, I found that with the help of these tools, I could quickly create, deploy, and manage applications while also enjoying enhanced security, portability, and scalability.

Network Configuration

As a cybersecurity professional, one of the most important capabilities that I should have is the ability to configure and manage network settings on Linux systems. It is through the mastery of these settings that I can quickly set up testing environments, manage network traffic, and identify or exploit different vulnerabilities. I also found out that knowing Linux network configuration very well allows me to adapt my testing approaches to the specific needs of the clients, which in turn, optimizes my overall workflows and the results obtained.

Remote Desktop Protocols in Linux

I learned of the two major protocols in Linux: the Remote Desktop protocol and the Virtual Network Computing.

Linux Security

All computer systems have a built-in intrusion risk and some, such as web servers facing the internet and running sophisticated applications, are more vulnerable than others. While it is decided that Linux systems are not as much exposed to Windows-specific viruses as well as the attack surface is not as large as in the case of Active Directory domain-joined hosts, still, it is extremely important to carry out basic security measures. Besides, I learned that the very same core security practices must be applied to every Linux environment, no matter what role it plays, in order to ensure its security.

Firewall Setup

I learned that the firewalls' primary purpose is to be a security barrier that monitors and controls the traffic between different network parts, like the interior and exterior zones. I came to the conclusion that they are indispensable in protecting networks from being infiltrated by unauthorized users and preventing harmful activities. Additionally, I found out that Linux has integrated firewall functions that enable me to manage the traffic coming and going to the network. Ultimately, I learned that through the application of rules set in advance according to

protocols, ports, and other factors, the unauthorized access can be successfully blocked and the security threats can be minimized.

System Logs

I came to recall that the Linux system logs are files that log activities of the system and they are major instruments for the supervision of the performance and the solving of the issues. I have learned that the security weaknesses can be discovered by these logs which are providing insights into the behavior of the network, application activity, and the user actions. It also learned that monitoring and analyzing the data help me to notice the unusual patterns like the unauthorized login or clear-text credentials which in turn lead to detection of the possible security violations.

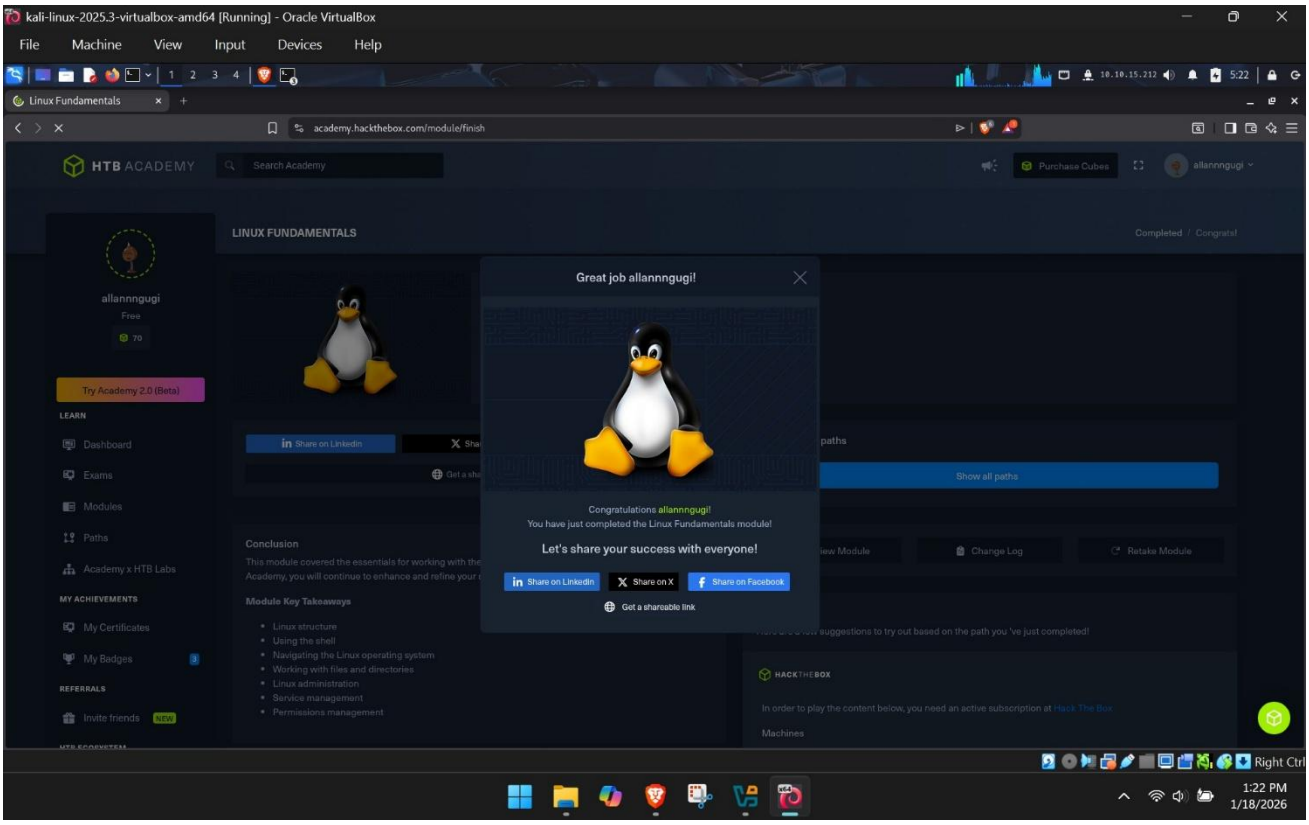
Solaris

I discovered that Solaris is a Unix-like Operating System created by Sun Microsystems in the 1990s and later acquired by Oracle. It is still extremely praised today for its excellent performance, high scalability, and support for advanced hardware and software configurations. I came across the fact that it is a very common choice in the enterprise sector, where it handles essential operations such as database management, cloud, and virtualization, among others.

Shortcuts

This section covered the shortcuts that make one’s life easier when using Linux.

Module Completion Screenshot



Conclusion

Looking back on this journey, I realize that my understanding of Linux and security has expanded considerably. Practicing applications such as curl and systemctl was a great way to bridge the theory I had learned and the actual practice, as a result I became more confident in the terminal. Although I faced difficulty at some points with filtering and interpreting the complicated data, overcoming those difficulties advanced my problem-solving skills and brought me to be more tolerant as well. I can truly say that I have been curious all along the way and I am going to learn with a sharper, more analytical perspective.